

Ahmed Asl Portfolio - Codebase & Documentation PDF

Bundled for direct upload to Kimi / Moonshot AI

Table of Files Included:

- AGENTS.md
- CLAUDE.md
- EASE-REPORT.md
- README.md
- app\about\page.js
- app\contact\page.js
- app\game-theme.css
- app\globals.css
- app\icon.svg
- app\layout.js
- app\not-found.js
- app\page.js
- app\robots.js
- app\series-theme.css
- app\sitemap.js
- app\work\page.js
- app\writing\[slug]\page.js
- app\writing\page.js
- components\ArticleEnhancer.js
- components\JsonLd.js
- components\MotionSystem.js
- components\PixelWorld.js
- components\PostCard.js
- components\ProjectCard.js
- components\SectionHeading.js
- components\SiteFooter.js
- components\SiteHeader.js
- components\SocialIcon.js
- components\WorldGallery.js
- components\WritingIndex.js
- components\WritingSeries.js
- content\templates\bandit.md
- content\templates\case-study.md
- content\templates\cybersecurity.md
- content\templates\embedded.md
- content\templates\flutter.md
- content\templates\learn-dart-flutter.md
- content\templates\learn-git.md
- content\templates\learn-linux.md
- content\templates\learn-python.md
- content\templates\linux.md
- content\templates\model.md
- content\templates\note.md
- content\writing\overthewire-bandit-workflow.md
- content\writing\reliable-sensor-to-flutter-pipeline.md
- content\writing\welcome-to-field-notes.md
- data\ambient.js
- data\portfolio.js
- data\publications.json
- docs\superpowers\specs\2026-07-25-mission-motion-system-design.md
- docs\superpowers\specs\2026-07-25-portfolio-narrative-design.md

- jsconfig.json
- lib\content.js
- lib\seo.js
- lib\site.js
- next.config.mjs
- package.json
- scripts\boost-web-sharpness.py
- scripts\bundle-for-kimi.py
- scripts\finalize-agribot.py
- scripts\fix-run-artifact.mjs
- scripts\new-post.mjs
- scripts\optimize-existing-assets.py
- scripts\optimize-repo-images.py
- scripts\prepare-ease.mjs
- scripts\studio.mjs
- scripts\sync-scholar-publications.mjs
- scripts\validate-content.mjs

File (1/30): AGENTS.md

```
<!-- BEGIN:nextjs-agent-rules -->
# This is NOT the Next.js you know
```

This version has breaking changes – APIs, conventions, and file structure may all differ from your training data. Read the relevant guide.

```
<!-- END:nextjs-agent-rules -->
```

File (2/30): CLAUDE.md

@AGENTS.md

File (3/30): EASE-REPORT.md

EASE Report

```
- Run: `ahmed-asl-portfolio-20260720`
- Source: `https://github.com/ahmed-ibrahim-asl/myPortfolio`
- Delivery target: `D:\work\portfliWebsite\myPortfolio`
- Approved and actual external media spend: `$0.00`
- Automated image QA: `5 passed / 0 failed`
- Production build: passed in normal and GitHub Pages `/myPortfolio` modes
- Content validation: 3 Markdown articles passed
- Dependency audit: 0 vulnerabilities
- Live local checks: 13 site, asset, and Studio responses returned HTTP 200
- Studio workflow: create, save, render, upload, read-back, and cleanup passed
- Deployment: not performed because the user did not request a push or deploy
- Responsive browser visual QA: blocked by the environment's `node_repl` Windows sandbox helper failure after the required troubleshooting
```

Stage status

```
- 01 Preflight - complete
- 02 Crawl - complete
- 03 Brand normalization - complete
- 04 Design system - complete
- 05 Media planning - complete
- 06 Prompt compilation - complete; no generated media needed
- 07 Image manifest - complete
- 08 Model routing - complete; local existing assets selected
- 09 Spend authorization - complete; no spend required
- 10 Asset acquisition - complete
- 11 Asset optimization and QA - complete
- 12 Artifact packaging - complete
- 13 Next.js implementation - complete
- 14 Build and automated verification - complete
- 15 Serve, deploy, and visual verification - partial; local serving/report complete, deployment not requested, browser visual inspection pending
```

Selected media

```
- PASS `identity-profile` - verified repository portrait, optimized WebP
- PASS `project-agribot` - verified repository image, optimized WebP
- PASS `project-rov` - verified repository image, optimized WebP
- PASS `project-security-lock` - verified repository image, optimized WebP
- PASS `project-megasumo` - verified repository image, optimized WebP
```

Design decisions

```
- The requested 0.80 scale is implemented as a compact spacing and type system, not CSS `zoom`.
- The public site uses solid warm-paper and ink surfaces with copper accents; it contains no CSS gradients.
- Markdown files remain portable and are the source of truth for blog content.
- Portfolio Studio is local-only; publishing uses Git and the existing GitHub Pages workflow.
```

Motion implementation

```
- FluxUI lane: `technical-systems`; moderate 240-400 ms transitions with restrained 55 ms staggering.
- Added viewport reveals, grid entrances, precise border draws, hero line choreography, schematic SVG tracing, reading progress, mobile-optimized.
- Continuous ambient animation is intentionally absent; the availability pulse stops after two cycles.
- `prefers-reduced-motion` reveals content immediately and disables decorative transforms and animation.
```

Pixel-game redesign

```
- Reference evidence: bounded public crawl of the supplied OverTheWire article plus the supplied pixel-noir screenshot.
- Visual direction: original engineer-RPG world, player profile, mission archive, skill tree, field logs, quest-index walkthroughs, and
```

- UI/UX Pro: Arcade & Retro Game advisor run at variance 8, motion 6, density 7; 44px targets, visible focus, contrast, responsive
- Deliberate overrides: generic rounded bento cards, external web-font dependency, glassmorphism, gradients, neon glare, copied artwork,
- Verification: GitHub Pages production build passed, three articles validated, and six public routes returned HTTP 200 locally.
- Publication: not performed; all work remains local.

Verified source-content and animation expansion

- Crawled the deployed portfolio HTML at `https://ahmed-ibrahim-asl.github.io/myPortfolio/`; no screenshots were used as content evidence
- Restored prominent CV access, two professional experience records, the Google Scholar publication, and six Technical Tutorials & Workshops
- Verified all five user-supplied Hwangstice GIF URLs as `image/gif`. They remain remote, visibly credited embeds because no reuse licenses
- Researched two more Hwangstice covers but did not select them because the same licensing uncertainty applies.
- Added a 1024 x 512, 24-frame OpenGameArt animated cityscape under CC0 1.0 as the reusable expansion asset.
- Added a user-controlled animation pause/play state that defaults to paused under `prefers-reduced-motion`.
- Removed elevated contrast filters from both portrait uses and softened project/workshop photo treatment; pixel rendering is applied on
- Production build and three-article validation passed after the expansion.
- Browser screenshot QA remains blocked by the Windows sandbox helper; deterministic local route and asset checks were used.
- Publication was not performed; all changes remain local.

Repository image sync

- Source: `https://github.com/ahmed-ibrahim-asl/myPortfolio/tree/main/images`
- Imported 14 of 14 files from the verified `origin/main` tree into `public/images`.
- Every local file matches its GitHub blob SHA exactly and decodes successfully.
- All 14 assets are used: six project primaries, two security-lock detail images, three workshop covers, and three profile portraits.
- Original JPEG/PNG encodings are retained. No CSS contrast, saturation, brightness, grayscale, forced pixel rendering, or image zoom is
- The homepage and About primary portrait now use `images/profilePicture/profile3.png`.
- The Work page exposes the repository robotics and security evidence; the About page exposes all three profile portraits and the three
- Detailed dimensions, byte sizes, source URLs, hashes, and usage locations are recorded in `.ease/repository-image-inventory.json`.
- No repository push or deployment was performed.

2026-07-24 About and research enhancement

- Replaced the flat technology tag cloud with three system-oriented capability groups.
- Added the four user-verified courses taught at Delta University.
- Replaced the outdated single publication record with a two-item cached Google Scholar feed.
- Added `npm run sync:scholar` and a weekly, credential-gated workflow using the Google Scholar Author adapter.
- Production build passed and `/` plus `/about/` returned HTTP 200 locally.
- No deployment or GitHub push was performed.

2026-07-24 Writing routes and identity links

- Added a dedicated OverTheWire Bandit walkthrough route group that surfaces the existing article as an unpublished draft without export
- Added a Learn With Me series with Linux, Python, Dart / Flutter, and Git tracks; empty tracks use explicit queued states instead of in
- Added five matching Markdown templates to Portfolio Studio so new posts inherit the correct series, category, and tags.
- Classified the Scientific Reports ROUV article as a Q1 Journal Paper and the U-Net aerial segmentation article as a Conference Paper.
- Preserved publication classification fields in the Scholar synchronization adapter so a refresh cannot silently erase them.
- Added bundled vector marks and visible labels for GitHub, LinkedIn, Google Scholar, TryHackMe, Behance, and YouTube in the footer and
- Content validation and the production static build passed; `/`, `/writing/`, `/about/`, and `/contact/` returned HTTP 200 locally.
- Rendered HTML checks found 6 footer icons, 12 Contact-page icons including the footer, both publication classifications, all five writ
- Responsive breakpoint and overflow safeguards were verified statically. The in-app visual browser remained unavailable because the Win
- No deployment, GitHub push, commit, or remote content publication was performed.

2026-07-25 Site Polish copy and SEO pass

- Rewrote the homepage, Work, About, Writing, Contact, article template, project summaries, expertise descriptions, tutorials, and the p
- Removed the unsupported AgriBot outcome claim "Published robotics research" and replaced it with a factual system-integration outcome.
- Added unique page titles, meta descriptions, canonical URLs, author metadata, Open Graph fields, Twitter cards, and a generated 1200 x
- Added Person and WebSite JSON-LD globally, ProfilePage JSON-LD on About, and BlogPosting JSON-LD with author, dates, keywords, and ima
- Added a pixel-style SVG favicon and `public/llms.txt` for a concise machine-readable portfolio summary.
- Stabilized sitemap modification dates for static pages and confirmed that draft articles remain absent from the export and sitemap.
- Content validation passed for three Markdown articles. Normal and GitHub Pages `/myPortfolio` production builds passed.
- Export assertions passed for six public HTML routes: one H1 each, unique descriptions, expected titles and canonicals, social metadata
- The local HTML routes, crawl files, `llms.txt`, both social cards, and the profile image returned HTTP 200; the extensionless social-c
- Scholar sync was checked and safely skipped because `SERPAPI\_KEY` is not configured; the verified local publication cache remains acti
- No deployment, GitHub push, commit, or remote content publication was performed.

2026-07-25 Mission Motion System and Portfolio Completion

- Enacted the approved Mission Motion System specification (`docs/superpowers/specs/2026-07-25-mission-motion-system-design.md`) and nar
- Attached stable `data-mission` attributes to all 8 homepage sections: `origin`, `credentials`, `projects`, `method`, `toolkit`, `story
- Upgraded `MotionSystem` with a unified IntersectionObserver strategy that computes active mission state (`data-active-mission`), track
- Implemented `MissionRail`: a fixed desktop navigation rail showing mission numbers (`01.08`), labels, active states, and completed st
- Added SVG `SignalPath` to the Method section to trace Question -> Learn -> Build -> Test upon activation.
- Added Toolkit system node activation styling and timing tokens (`--motion-instant` through `--motion-scene`).
- Verified zero em dash characters (`-`) in visible site copy and zero unsupported academic claims or invented statistics.
- Verified exact education string: `Postgraduate Qualifying Studies in Mechatronics, Mansoura University, February 2025 to Present`.
- Refined `MissionRail` display rules so the fixed right panel renders only at ultra-wide viewports (`>= 1440px`), preventing any con
- Thoroughly overhauled the About page (`/about/`) responsive grid system across `about-intro`, `about-statement`, `profile-credential-c
- Removed all leftover light theme paper variables (`--paper: #f2f0e9`, `--paper-deep: #e7e3d8`) and light body backgrounds from `global

- Packaged all site source files, components, data, studio, assets, and documentation into a single clean local zip archive: `portfolio\_`
- Content validation passed (`npm run validate:content`).
- Production builds passed in both standard and GitHub Pages `/myPortfolio` static export modes.
- `git diff --check` passed cleanly without whitespace issues.
- All work remains strictly local; no push, upload, or deployment was performed.

File (4/30): README.md

Ahmed Asl – Portfolio & Field Notes

A compact, content-first engineering portfolio and technical writing platform. The public website is a statically exported Next.js app. Blog posts are ordinary Markdown files, managed through the included local Portfolio Studio.

Run the website

```
```bash
npm ci
npm run dev
```
```

Open `http://localhost:3000`.

Update the blog with Portfolio Studio

```
```bash
npm run studio
```
```

Open `http://localhost:4173`, then:

1. Select **Create article** and choose a reusable template.
2. Write in Markdown, use **Split preview**, and upload article images if needed.
3. Select **Save draft** to write the post to your computer only.
4. Select **Publish to GitHub** when it is ready for the public website.

There is no database and no hosted admin panel. The repository is the content system:

- **Save draft** writes `content/writing/<slug>.md`.
- **Upload image** writes `public/blog/<slug>/<image>` and inserts its Markdown.
- **Publish to GitHub** changes `draft` to `false`, commits that article and its image folder, then runs `git push`.
- The workflow in `.github/workflows/deploy-pages.yml` builds the updated static site and GitHub Pages replaces the live version.

Before using the Publish button for the first time, commit and push this site redesign once so the remote `main` branch contains the new application. Git must also already be authenticated on the computer running Studio.

Add a post from the terminal

```
```bash
npm run new:post -- --type=cybersecurity --title="OverTheWire Bandit"
```
```

Available templates: `cybersecurity`, `linux`, `flutter`, `embedded`, `model`, `case-study`, and `note`.

Content map

- `content/writing/\*.md` – published articles and drafts
- `content/templates/\*.md` – reusable writing structures
- `public/blog/<slug>/` – article images
- `data/portfolio.js` – projects, experience, tutorials, and contact data
- `studio/` and `scripts/studio.mjs` – the local editor

Verify a change

```
```bash
npm run validate:content
npm run build
```
```

The static output is written to `out/`.

```
## Refresh Google Scholar publications
```

The public website reads `data/publications.json`, so it never exposes an API key and remains compatible with static GitHub Pages hosting. The sync command uses the SerpApi Google Scholar Author adapter instead of scraping Scholar in a visitor's browser.

```
```powershell
Copy-Item .env.example .env
npm run sync:scholar
```
```

Set `SERPAPI\_KEY` in `.env` before running the command. Without it, the command keeps the verified local cache unchanged. For automatic weekly updates, add the same value as the repository secret `SERPAPI\_KEY`; the local workflow `.github/workflows/sync-scholar.yml` refreshes the cache and only commits when the publication records change.

File (5/30): `about\page.js`

```
import { JsonLd } from "@components/JsonLd";
import {
  coursesTaught,
  education,
  experience,
  profile,
  publications,
  publicationSource,
  technologyGroups,
  toolkitHeading,
  toolkitIntro,
  tutorials
} from "@data/portfolio";
import { createPageMetadata, createProfilePageJsonLd } from "@lib/seo";

export const metadata = createPageMetadata({
  title: "Embedded Systems Engineer and Educator",
  description:
    "Meet Ahmed Asl, an embedded systems and IoT engineer and teaching assistant. View his experience, technical skills, publications, and more.",
  pathname: "/about/"
});

export default function AboutPage() {
  return (
    <div>
      <JsonLd data={createProfilePageJsonLd()} />

      <section className="page-intro shell about-intro">
        <div>
          <p className="eyebrow">About Ahmed Asl</p>
          <h1>Titles tell you where someone works. Questions tell you how they think.</h1>
          <p className="page-lede">
            Mine started with an ATM. I wanted to know how it recognized an account,
            checked whether money was available, and exchanged information with systems
            somewhere else.
          </p>
        </div>
        <figure className="about-portrait">
          <img src={profile.portrait} alt={`Portrait of ${profile.name}`} />
        </figure>
      </section>

      <section className="shell about-statement about-statement-focused">
        <div className="about-narrative">
          <p>
            That curiosity moved into Windows CMD, BIOS passwords, online games,
            electronics, networks, and eventually the machines and connected products
            I build today.
          </p>
          <p>
            I do not see firmware, electronics, Linux, security, Flutter, or design as
            separate identities. They are tools I learned while building projects and
            trying to solve problems. I keep sharpening them because the next problem
            rarely stays inside one field.
          </p>
        </div>
      </section>
    </div>
  );
}
```

```

<lt;p>
    That is what I bring to a project: I stay with the problem, learn what is
    missing, test what I build, and make the result understandable.
<lt;/p>
<lt;/div>
<lt;aside className="profile-credential-card">
    <lt;div>
        <lt;p className="eyebrow">Curriculum vitae</p>
        <lt;h2>Education, experience, and technical record</h2>
        <lt;p>
            Open my CV for degree, work history, projects, courses, and
            technical skills.
        </p>
    </div>
    <lt;a className="button primary" href={profile.cv} target="_blank" rel="noreferrer">
        Open CV <span aria-hidden="true">▶</span>
    </a>
</aside>
</section>

<lt;section id="experience" className="section section-ink">
    <lt;div className="shell">
        <lt;div className="section-heading">
            <lt;div>
                <lt;p className="eyebrow">Experience</p>
                <lt;h2>Engineering experience and university teaching</h2>
            </div>
        </div>

        <lt;div className="timeline">
            {experience.map((item, index) => (
                <lt;article className="timeline-item" key={`-${item.role}-${item.period}`}>
                    <lt;span className="timeline-index mono">
                        {String(index + 1).padStart(2, "0")}
                    </span>
                    <lt;div>
                        <lt;h3>{item.role}</h3>
                        <lt;p className="timeline-org">{item.organization}</p>
                        <lt;p className="timeline-type mono">{item.type}</p>
                    </div>
                    <lt;div>
                        <lt;p>{item.description}</p>
                        <lt;div className="tag-row dark">
                            {item.tags.map((tag) => (
                                <lt;span className="tag" key={tag}>{tag}</span>
                            ))}
                        </div>
                    </div>
                    <lt;div className="timeline-meta mono">
                        <lt;span>{item.period}</span>
                        <lt;span>{item.location}</span>
                    </div>
                </article>
            ))}

            {/* Render Education from the data layer */}
            {education.map((item, index) => (
                <lt;article className="timeline-item" key={`edu-${index}`}>
                    <lt;span className="timeline-index mono">
                        {String(experience.length + index + 1).padStart(2, "0")}
                    </span>
                    <lt;div>
                        <lt;h3>{item.institution}</h3>
                        <lt;p className="timeline-org">Education</p>
                    </div>
                    <lt;div>
                        <lt;p>{item.credential}</p>
                    </div>
                    <lt;div className="timeline-meta mono">
                        <lt;span>{item.period}</span>
                    </div>
                </article>
            ))}
        </div>
    </div>
</section>

<lt;section id="teaching" className="section teaching-section">

```

```

<div className="shell">
  <div className="section-heading">
    <div>
      <p className="eyebrow">Teaching portfolio</p>
      <h2>Courses taught</h2>
    </div>
  </div>
  <div className="courses-grid">
    {coursesTaught.map((course, index) => (
      <article className="course-card" key={course}>
        <div className="course-card-top mono">
          <span>COURSE_{String(index + 1).padStart(2, "0")}</span>
          <span>Delta University</span>
        </div>
        <h3>{course}</h3>
        <p>Undergraduate lectures and laboratory instruction.</p>
      </article>
    ))}
  </div>
</div>
</section>

<section id="skills" className="section shell about-capabilities-section">
  <div className="section-heading">
    <div>
      <p className="eyebrow">Engineering toolkit</p>
      <h2>{toolkitHeading}</h2>
      <p className="section-intro">
        {toolkitIntro}
      </p>
    </div>
  </div>
  <div className="technology-matrix">
    {technologyGroups.map((group) => (
      <article className="technology-cluster" key={group.index}>
        <div className="technology-cluster-top">
          <span className="technology-cluster-index mono">{group.index}</span>
          <span className="mono">SYSTEM CLASS</span>
        </div>
        <h3>{group.title}</h3>
        <p>{group.description}</p>
        <ul className="technology-tool-list" aria-label={` ${group.title} tools`>
          {group.tools.map((tool, index) => (
            <li key={tool}>
              <span className="mono" aria-hidden="true">
                {String(index + 1).padStart(2, "0")}
              </span>
              <strong>{tool}</strong>
            </li>
          ))}
        </ul>
      </article>
    ))}
  </div>
</section>

<section id="publications" className="section section-ink research-record-section">
  <div className="shell research-feed">
    <div className="section-heading research-feed-heading">
      <div>
        <p className="eyebrow">Research record</p>
        <h2>Publications</h2>
        <p className="section-intro">
          Journal and conference papers from my Google Scholar profile,
          with publication type, ranking, year, and citation count.
        </p>
      </div>
      <a
        className="text-link"
        href={publicationSource.profileUrl}
        target="_blank"
        rel="noreferrer"
      >
        Open Google Scholar <span aria-hidden="true">█</span>
      </a>
    </div>
    <div className="publication-list">
      {publications.map((item, index) => (

```

```

<article className="publication-record" key={item.id}>
  <div className="publication-record-index mono">
    <span>PUBLICATION_{String(index + 1).padStart(2, "0")}</span>
    <strong>{item.year}</strong>
  </div>
  <div className="publication-record-copy">
    <div
      className="publication-classification"
      aria-label="Publication classification"
    >
      {item.ranking ? (
        <span className="publication-ranking">{item.ranking}</span>
      ) : null}
      <span className="publication-type">
        {item.publicationType || "Publication"}
      </span>
    </div>
    <h3>
      <a href={item.href} target="_blank" rel="noreferrer">
        {item.title}
      </a>
    </h3>
    <p className="publication-authors">{item.authors}</p>
    <p className="publication-venue">{item.venue}</p>
    <div className="tag-row dark">
      {item.tags.map((tag) => (
        <span className="tag" key={tag}>{tag}</span>
      ))}
    </div>
  </div>
  <div className="publication-record-meta mono">
    <span>
      {item.citedBy} {item.citedBy === 1 ? "citation" : "citations"}
    </span>
    <a href={item.href} target="_blank" rel="noreferrer">
      Open publication <span aria-hidden="true">■</span>
    </a>
  </div>
</article>
  ))}
</div>
<p className="publication-sync-note">
  Each record links to the paper or its Google Scholar entry.
</p>
</div>
</section>

<section id="tutorials" className="section shell">
  <div className="section-heading">
    <div>
      <p className="eyebrow">Teaching resources</p>
      <h2>Technical Tutorials & Workshops</h2>
      <p className="section-intro">
        Recorded lessons and workshops on ROS, embedded systems, digital
        logic, Matlab, and networking.
      </p>
    </div>
  </div>
  <div className="tutorial-grid">
    {tutorials.map((tutorial) => (
      <a className="tutorial-card" href={tutorial.href} target="_blank" rel="noreferrer" key={tutorial.title}>
        <img src={tutorial.image} alt={` ${tutorial.title} cover`} loading="lazy" />
      </div>
      <div className="tag-row">
        {tutorial.tags.map((tag) => (
          <span className="tag" key={tag}>{tag}</span>
        ))}
      </div>
      <h3>{tutorial.title}</h3>
      <p>{tutorial.description}</p>
      <span className="text-link">Watch session <span aria-hidden="true">■</span></span>
    </div>
    </a>
  ))}
</div>
</section>

/* Client CTA at bottom as per spec */

```

```

    <section className="section home-contact-section">
    <div className="shell home-contact-grid">
      <div>
        <p className="eyebrow">Contact</p>
        <h2>Bring me the problem, even if the solution is not clear yet.</h2>
        <p>
          Tell me what you are trying to build or fix, what already exists, and where
          you are stuck. I will tell you whether I can help and what the next useful
          step should be.
        </p>
      </div>
    </div>
  ... [Truncated 13 lines in PDF preview - see TXT bundle for full text]

```

File (6/30): app\contact\page.js

```

import { SocialIcon } from "@components/SocialIcon";
import { profile } from "@data/portfolio";
import { createPageMetadata } from "@lib/seo";

export const metadata = createPageMetadata({
  title: "Contact for Hardware and IoT Projects",
  description:
    "Contact Ahmed Asl about hardware prototypes, IoT products, embedded systems, robotics, connected products, mechatronics, or engineering projects.",
  pathname: "/contact/"
});

export default function ContactPage() {
  return (
    <section className="contact-page shell">
      <div className="contact-lead">
        <p className="eyebrow">Contact / Engineering collaboration</p>
        <h1>Bring me the problem, even if the solution is not clear yet.</h1>
        <p>
          Tell me what you are trying to build or fix, what already exists, and where
          you are stuck. I will tell you whether I can help and what the next useful
          step should be.
        </p>
      </div>

      <div className="contact-grid">
        <form
          className="contact-form"
          action={`https://formsubmit.co/${profile.email}`}
          method="POST"
        >
          <input type="hidden" name="_subject" value="New portfolio inquiry" />
          <input type="hidden" name="_captcha" value="false" />
          <label>
            <span>Name or organization</span>
            <input name="name" required placeholder="Your name" />
          </label>
          <label>
            <span>Email address</span>
            <input name="email" type="email" required placeholder="you@example.com" />
          </label>
          <label>
            <span>Project brief</span>
            <textarea
              name="details"
              required
              rows="8"
              placeholder="Describe the system, constraints, current state, and desired result."
            />
          </label>
          <button className="button primary" type="submit">
            Send your project brief <span aria-hidden="true">📧</span>
          </button>
        </form>

        <aside className="contact-aside">
          <div>
            <p className="eyebrow">Direct contact</p>
            <div className="contact-socials">
              <a href={`mailto:${profile.email}`} aria-label="Email Ahmed directly">
                <span className="social-link-text">Email directly</span>

```

```

        <span className="social-link-arrow" aria-hidden="true">■</span>;
    </a>;
    <a href={profile.whatsapp} target="_blank" rel="noreferrer" aria-label="Message Ahmed on WhatsApp">
        <span className="social-link-text">Message on WhatsApp</span>;
        <span className="social-link-arrow" aria-hidden="true">■</span>;
    </a>;
</div>;
</div>;
</div>;
<p className="eyebrow">Evaluation resources</p>;
<div className="contact-socials">
    <a href={profile.cv} target="_blank" rel="noreferrer" aria-label="Open CV for employment evaluation">
        <span className="social-link-text">Open CV</span>;
        <span className="social-link-arrow" aria-hidden="true">■</span>;
    </a>;
    <a href={profile.scholar} target="_blank" rel="noreferrer" aria-label="Open Google Scholar for academic evaluation">
        <span className="social-link-text">Open Publications</span>;
        <span className="social-link-arrow" aria-hidden="true">■</span>;
    </a>;
</div>;
</div>;
</div>;
<p className="eyebrow">Professional profiles</p>;
<div className="contact-socials">
    {profile.socials.map((social) => {
        <a
            key={social.label}
            href={social.href}
            target="_blank"
            rel="noreferrer"
            aria-label={` ${social.label} profile (opens in a new tab)`}
        >;
        <SocialIcon label={social.label} />;
        <span className="social-link-text">{social.label}</span>;
        <span className="social-link-arrow" aria-hidden="true">■</span>;
    </a>;
    )}}
</div>;
</div>;
<p className="availability-note">
    <span className="status-dot" aria-hidden="true" />
    {profile.availability}
</p>;
</aside>;
</div>;
</section>;
);
}

```

File (7/30): app\game-theme.css

```

/* Pixel-noir portfolio theme: flat colors, crisp edges, readable long-form content. */

:root {
  color-scheme: dark;
  --paper: #060814; --paper-deep: #0b0f21; --ink: #eef1ff; --ink-soft: #c6cdea;
  --muted: #9ca7ca; --line: #30395e; --signal: #55d5d8; --signal-dark: #55d5d8;
  --night: #050713; --night-soft: #10152c; --night-line: #30395e; --night-text: #eef1ff;
  --panel: #0c1023; --panel-raised: #121831; --pixel-cyan: #55d5d8;
  --pixel-green: #8edb7a; --pixel-gold: #f0c66c; --pixel-shadow: #02030a;
  --radius: 0; --header-height: 68px;
  --font-ui: "Cascadia Code", "SFMono-Regular", Consolas, monospace;
  --font-editorial: "Cascadia Code", "SFMono-Regular", Consolas, monospace;
  --font-mono: "Cascadia Code", "SFMono-Regular", Consolas, monospace;
  --font-reading: "Segoe UI", Arial, sans-serif;

  /* Mission motion timing tokens */
  --motion-instant: 140ms;
  --motion-fast: 180ms;
  --motion-standard: 280ms;
  --motion-section: 420ms;
  --motion-scene: 520ms;
  --motion-stagger: 45ms;
}

```

```

html { overflow-x: hidden; color-scheme: dark; background: var(--night); }
body { min-width: 320px; overflow-x: hidden; color: var(--ink); background: var(--night); font-family: var(--font-ui); font-size: 15px; }
img { filter: none !important; image-rendering: auto !important; }
body::before { z-index: -1; border: 8px solid #090c1d; box-shadow: inset 0 0 0 1px #1a203d; }
body::after {
  content: "";
  position: fixed;
  inset: 0;
  z-index: -1;
  background-image: linear-gradient(to right, rgba(48, 57, 94, 0.07) 1px, transparent 1px),
    linear-gradient(to bottom, rgba(48, 57, 94, 0.07) 1px, transparent 1px);
  background-size: 32px 32px;
  pointer-events: none;
}
::selection { color: var(--night); background: var(--pixel-cyan); }
a, button { cursor: pointer; }
a, button, input, textarea { border-radius: 0; }
h1, h2, h3 { color: var(--ink); font-family: var(--font-editorial); font-weight: 800; letter-spacing: -0.045em; text-wrap: balance; }
h1 { line-height: .95; text-transform: uppercase; }
h2 { line-height: 1.05; }
p { color: var(--ink-soft); }
.shell { width: min(calc(100% - 48px), var(--shell)); }
.section { padding-block: 88px; }
.page-section { padding-block: 28px 92px; }
.skip-link { z-index: 60; color: var(--night); background: var(--pixel-cyan); border: 2px solid var(--ink); box-shadow: 4px 4px 0 var(--ink); }
.muted { color: var(--muted); }
.eyebrow { display: inline-flex; gap: 8px; align-items: center; margin-bottom: 14px; color: var(--pixel-cyan); font-size: .72rem; font-weight: 800; }
.eyebrow::before { width: 10px; height: 10px; content: ""; background: var(--pixel-cyan); box-shadow: 4px 0 0 #263051; }

.site-header { z-index: 40; height: var(--header-height); background: #070a18; border-top: 4px solid #1d2340; border-bottom: 2px solid var(--ink); }
.header-inner { gap: 24px; }
.brand { gap: 12px; min-height: 44px; color: var(--ink); font-size: .76rem; font-weight: 800; letter-spacing: .08em; }
.brand-mark { display: grid; width: 34px; height: 30px; color: var(--night); background: var(--pixel-cyan); border: 2px solid #baf6f2; border-radius: 4px; }
.brand:hover .brand-mark, .brand:focus-visible .brand-mark { color: var(--night); background: var(--pixel-cyan); transform: translate(2px, 2px); }
.site-nav { display: flex; gap: 4px; align-items: center; }
.site-nav a { position: relative; display: inline-flex; align-items: center; min-height: 44px; padding: 0 13px; color: var(--muted); font-weight: 800; }
.site-nav a:hover, .site-nav a:focus-visible, .site-nav a.active { color: var(--ink); background: var(--panel-raised); border-color: var(--ink); }
.site-nav a.active::after { position: absolute; right: 8px; bottom: 4px; left: 8px; height: 3px; content: ""; background: var(--pixel-cyan); }
.menu-toggle { display: none; min-height: 44px; color: var(--ink); background: var(--panel); border: 1px solid var(--line); }
.scroll-progress { z-index: 50; height: 4px; background: #161c38; }
.scroll-progress span { background: var(--pixel-cyan); }

.hero { position: relative; padding: 26px 0 24px; background: #060814; border-bottom: 2px solid var(--line); }
.hero-shell { position: relative; display: grid; }
.pixel-world { position: relative; grid-area: 1/1; min-height: 600px; overflow: hidden; background: #080b1b; border: 2px solid #3a446d; }
.pixel-world::before, .pixel-world::after { position: absolute; z-index: 2; width: 22px; height: 22px; pointer-events: none; content: ""; }
.pixel-world::before { top: 10px; left: 10px; border-top: 4px solid var(--pixel-cyan); border-left: 4px solid var(--pixel-cyan); }
.pixel-world::after { right: 10px; bottom: 10px; border-right: 4px solid var(--pixel-cyan); border-bottom: 4px solid var(--pixel-cyan); }
.pixel-world svg { display: block; width: 100%; min-width: 760px; height: 100%; min-height: 360px; }
.pixel-sky { fill: #080b1b; }
.pixel-stars { fill: #7783bd; animation: pixel-blink 520ms steps(2,end) 3; }
.pixel-clouds { fill: #151b35; }
.pixel-city-back { fill: #10142a; }
.pixel-windows { fill: #3d476e; }
.pixel-platforms { fill: #2a3151; }
.pixel-antenna { fill: #8d99c7; }
.pixel-bot-eye { fill: var(--pixel-green); }
.pixel-circuit { fill: #1b2342; stroke: var(--pixel-cyan); stroke-width: 6; stroke-dasharray: 12 8; stroke-dashoffset: 160; animation: circuit 4s linear infinite; }
.pixel-bot { fill: #8792bf; opacity: 0; transform: translateY(28px); animation: bot-spawn 340ms steps(4,end) 520ms forwards; }
.pixel-world-label { position: absolute; top: 14px; right: 28px; z-index: 10; padding: 5px 10px; color: #b9c1e2; background: #0a0d1d; border: 1px solid var(--ink); }
.hero-grid { position: relative; z-index: 4; display: grid; grid-area: 1/1; grid-template-columns: minmax(0,1fr) minmax(220px,320px); gap: 10px; }
.hero-copy { background: var(--panel); border: 2px solid #3a446d; box-shadow: 8px 8px 0 var(--pixel-shadow); }
.hero-copy { padding: clamp(26px,4vw,46px); }
.hero-portrait { width: 100%; padding: 8px; margin: 0; background: var(--panel); border: 2px solid #3a446d; box-shadow: 8px 8px 0 var(--pixel-shadow); }
.hero-portrait img { display: block; width: 100%; aspect-ratio: 4/5; object-fit: cover; object-position: center top; }
.hero-kicker { color: var(--pixel-green); font-size: .68rem; font-weight: 800; letter-spacing: .08em; text-transform: uppercase; }
.status-dot { width: 9px; height: 9px; background: var(--pixel-green); border-radius: 0; box-shadow: 3px 0 0 #25402f; }
.hero h1 { max-width: 760px; margin: 26px 0 22px; font-size: clamp(2.8rem,7.2vw,6rem); text-shadow: 5px 5px 0 #202746; }
.hero h1 span:last-child { color: var(--pixel-cyan); }
.hero-lede { max-width: 62ch; color: #c7cee8; font-family: var(--font-reading); font-size: clamp(1rem,1.7vw,1.2rem); }
.hero-actions { gap: 14px; margin-top: 30px; }
.button, .text-link, .filter-row button, .copy-code { font-family: var(--font-mono); font-weight: 800; text-transform: uppercase; }
.button { min-height: 46px; padding: 12px 16px; color: var(--ink); background: var(--panel-raised); border: 2px solid #4a557f; box-shadow: 4px 4px 0 var(--ink); }
.button.primary { color: #071018; background: var(--pixel-cyan); border-color: #b3f5f0; }
.button.light { color: #071018; background: var(--pixel-gold); border-color: #ffe4a6; }
.button:hover, .button:focus-visible { transform: translate(2px,2px); box-shadow: 2px 2px 0 var(--pixel-shadow); }
.button.text-button { color: var(--ink); background: #171d38; }
.hero-profile { position: relative; padding: 12px; }

```

```

.player-card-label { display: flex; align-items: center; min-height: 30px; padding: 0 8px; color: var(--pixel-gold); background: #181e39; }
.portrait-frame { padding: 8px 8px 36px; background: #090c1b; border: 0; }
.portrait-frame img { position: relative; z-index: 1; display: block; width: 100%; aspect-ratio: 4/5; object-fit: cover; object-position: center; }
.portrait-schematic { inset: -8px; z-index: 3; width: calc(100% + 16px); height: calc(100% + 16px); }
.portrait-schematic path { stroke: var(--pixel-cyan); }
.portrait-schematic circle { fill: var(--pixel-cyan); }
.portrait-index { right: auto; bottom: 9px; left: 9px; color: var(--muted); font-size: .58rem; }
.portrait-readout { top: 18px; right: -10px; color: #071018; background: var(--pixel-green); border-left: 4px solid #d8ffd0; font-size: .7rem; }
.profile-data > div { display: grid; grid-template-columns: 74px 1fr; gap: 10px; padding: 10px; background: #11162d; border: 0; border-bottom: 1px solid #090c1b; }
.profile-data > div:last-child { border-bottom: 0; }
.profile-data span { color: var(--pixel-cyan); font-size: .59rem; text-transform: uppercase; }
.profile-data strong { color: #dce1f6; font-size: .68rem; }
.technology-rail { display: flex; gap: 8px; padding-top: 26px; overflow-x: auto; }
.technology-rail span, .tag, .skills-list span { color: #bdc6e5; background: #11162c; border: 1px solid #394367; border-radius: 0; font-size: .7rem; }
.technology-rail span { flex: 0 0 auto; padding: 8px 11px; }
.section-heading { position: relative; align-items: end; padding-bottom: 20px; margin-bottom: 36px; border: 0; }
.section-heading::after { height: 2px; background: var(--line); }
.section-heading h2 { max-width: 780px; margin: 0; font-size: clamp(1.9rem, 4.4vw, 3.7rem); text-transform: uppercase; text-shadow: 3px 3px 0px #071018; }
.text-link { display: inline-flex; gap: 8px; align-items: center; min-height: 44px; color: var(--pixel-cyan); font-size: .7rem; }
.text-link:hover, .text-link:focus-visible { color: #b8f8f4; }

.project-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 26px; }
.project-card { overflow: visible; padding: 14px; color: var(--ink); background: var(--panel); border: 2px solid #354064; box-shadow: 7px 7px 0px #090c1b; }
.project-card::before { top: -2px; right: auto; left: 14px; width: 54px; height: 4px; background: var(--pixel-cyan); }
.project-card:hover, .project-card:focus-within { border-color: var(--pixel-cyan); box-shadow: 0 0 16px rgba(85, 213, 216, 0.22), 5px 5px 0px #090c1b; }
.project-card-top { padding: 2px 2px 10px; color: var(--muted); border: 0; }
.project-card-top span:first-child::before { content: "MISSION_"; }
.project-media { position: relative; aspect-ratio: 16/9; overflow: hidden; background: #151a32; border: 1px solid #465176; }
.project-media::before, .project-media::after { position: absolute; z-index: 2; width: 16px; height: 16px; pointer-events: none; content: ""; }
.project-media::before { top: 7px; left: 7px; border-top: 3px solid var(--pixel-cyan); border-left: 3px solid var(--pixel-cyan); }
.project-media::after { right: 7px; bottom: 7px; border-right: 3px solid var(--pixel-cyan); border-bottom: 3px solid var(--pixel-cyan); }
.project-media img { width: 100%; height: 100%; object-fit: cover; filter: none; image-rendering: auto; transition: filter 220ms ease, transform 220ms ease; }
.project-card:hover .project-media img { filter: none; transform: none; }
.project-placeholder { display: flex; align-items: end; justify-content: space-between; padding: 18px; color: var(--pixel-cyan); background: #11162d; }
.project-copy { padding: 20px 2px 16px; }
.project-category { color: var(--pixel-cyan); font-size: .67rem; font-weight: 800; letter-spacing: .08em; text-transform: uppercase; }
.project-copy h3 { margin-bottom: 12px; font-size: clamp(1.25rem, 2vw, 1.72rem); text-transform: uppercase; }
.project-copy > p:last-child, .expertise-item p, .post-main > p, .timeline-item p, .tutorial-card p { color: #adb6d5; font-family: monospace; }
.project-meta { padding: 14px 2px 2px; border-top: 1px solid #30395e; }
.tag-row { gap: 6px; }
.tag { padding: 5px 8px; }
.tag:hover { color: #071018; background: var(--pixel-cyan); border-color: #aff3ef; transform: translateY(-2px); }
.outcome { color: var(--pixel-green); font-size: .66rem; text-transform: uppercase; }

.section-ink { color: var(--ink); background: #090d1e; border-top: 2px solid #232a4b; border-bottom: 2px solid #232a4b; }
.section-ink::before, .publication-band::before, .site-footer::before { display: none; }
.section-ink .eyebrow { color: var(--pixel-cyan); }
.section-ink .section-heading h2 { color: var(--ink); }
.expertise-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); gap: 14px; }
.expertise-item { min-height: 244px; padding: 22px; background: var(--panel); border: 1px solid #354064; box-shadow: 4px 4px 0px var(--pixel-cyan); }
.expertise-item > .mono { display: inline-grid; width: 34px; height: 30px; color: #071018; background: var(--pixel-gold); place-items: center; }
.expertise-item h3 { margin: 34px 0 12px; color: var(--ink); font-size: 1.1rem; text-transform: uppercase; }
.expertise-item:hover { background: #151b35; transform: translate(2px, 2px); }

.post-list { display: grid; gap: 12px; }
.post-card { grid-template-columns: 58px minmax(0, 1fr) 52px; gap: 18px; padding: 22px; color: var(--ink); background: var(--panel); border: 1px solid #354064; }
.post-card:hover, .post-card:focus-within { background: #121831; border-color: var(--pixel-cyan); box-shadow: 0 0 16px rgba(85, 213, 216, 0.22), 5px 5px 0px #090c1b; }
.post-index { display: grid; width: 40px; height: 36px; color: #071018; background: var(--pixel-cyan); place-items: center; }
.post-index::before { content: "L"; }
.post-meta { color: var(--muted); font-size: .64rem; letter-spacing: .05em; text-transform: uppercase; }
.post-main h3 { margin: 8px 0 10px; font-size: clamp(1.16rem, 2.2vw, 1.56rem); text-transform: uppercase; }
.post-main h3 a:hover, .post-main h3 a:focus-visible { color: var(--pixel-cyan); }
.post-arrow { display: grid; min-width: 44px; min-height: 44px; color: var(--pixel-cyan); background: #171d38; border: 1px solid #3c476c; }

.publication-band { position: relative; color: var(--ink); background: #12162d; border-top: 2px solid #465176; border-bottom: 8px solid #090c1b; }
.publication-grid { gap: 48px; padding-block: 64px; }
.publication-grid h2 { color: var(--ink); font-size: clamp(1.7rem, 3.8vw, 3.2rem); text-transform: uppercase; }
.publication-grid > div:last-child { padding: 22px; background: #0b0f20; border: 1px solid #3b456a; box-shadow: 6px 6px 0px var(--pixel-cyan); }

.page-intro { display: grid; min-height: auto; padding-top: 82px; padding-bottom: 56px; border-bottom: 2px solid var(--line); }
.page-intro::before { display: block; width: 74px; height: 8px; margin-bottom: 28px; content: ""; background: var(--pixel-cyan); box-shadow: 4px 4px 0px #090c1b; }
.page-intro h1, .contact-lead h1, .not-found h1, .article-title-grid h1 { max-width: 980px; margin-bottom: 24px; font-size: clamp(2.5rem, 5.6vw, 3.2rem); }
.page-lede { max-width: 70ch; color: #b9c1dc; font-family: var(--font-reading); font-size: 1.08rem; line-height: 1.75; }
.archive-header { padding: 12px 14px; margin-bottom: 24px; color: var(--muted); background: var(--panel); border: 1px solid var(--line); }

.about-intro { display: grid; grid-template-columns: minmax(0, 1.2fr) minmax(240px, 340px); gap: 48px; align-items: center; }
.about-portrait { width: 100%; max-width: 340px; margin: 0 auto; padding: 10px; background: var(--panel); border: 2px solid #465176; box-shadow: 4px 4px 0px #090c1b; }

```

```

.about-portrait img { display: block; width: 100%; height: auto; aspect-ratio: 4/5; object-fit: cover; object-position: center top; filter: brightness(0.8); }
.about-statement { grid-template-columns: repeat(2,minmax(0,1fr)); gap: 36px; padding-block: 48px 72px; }
.about-statement p { padding: 22px 24px; margin: 0; color: #c4cbe4; background: var(--panel); border: 1px solid #354064; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.timeline { border-top: 0; }
.timeline-item { display: grid; grid-template-columns: 54px minmax(140px,.8fr) minmax(0,1.4fr) minmax(120px,.6fr); gap: 20px; padding: 20px 0; }
.timeline-index { display: grid; width: 38px; height: 34px; color: #071018; background: var(--pixel-gold); place-items: center; }
.timeline-item h3, .timeline-org { color: var(--ink); }
.timeline-meta { color: var(--pixel-cyan); }
.tutorial-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); gap: 18px; }
.tutorial-card { overflow: hidden; color: var(--ink); background: var(--panel); border: 1px solid #354064; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.tutorial-card > div { padding: 22px; }
.tutorial-card img { filter: none; image-rendering: auto; }
.tutorial-card: hover { border-color: #6875a8; transform: translateY(-3px); }
.tutorial-card h3 { color: var(--ink); text-transform: uppercase; }
.skills-index { padding: 26px; background: var(--panel); border: 2px solid #465176; box-shadow: 7px 7px 0 var(--pixel-shadow); }
.skills-list { gap: 8px; }
.skills-list span { padding: 8px 10px; }
.writing-tools { padding: 18px; margin-bottom: 26px; background: var(--panel); border: 1px solid #354064; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.search-field { min-height: 50px; color: var(--pixel-cyan); background: #070a18; border: 2px solid #3c476c; }
.search-field: focus-within { border-color: var(--pixel-cyan); box-shadow: 0 0 3px #1f3150; }
.search-field input { color: var(--ink); }
.search-field input::placeholder, .contact-form input::placeholder, .contact-form textarea::placeholder { color: #7782a8; }
.filter-row { gap: 8px; }
.filter-row button { min-height: 44px; padding: 8px 12px; color: var(--muted); background: #10152a; border: 1px solid #394367; font-size: 0.9em; }
.filter-row button: hover, .filter-row button: focus-visible, .filter-row button: active { color: #071018; background: var(--pixel-cyan); border: 1px solid #071018; }
.empty-state { padding: 44px; color: var(--ink); background: var(--panel); border: 2px dashed #465176; }

.article-page { padding-bottom: 80px; }
.article-header { padding-top: 70px; padding-bottom: 48px; border-bottom: 2px solid var(--line); }
.article-back { display: inline-flex; align-items: center; min-height: 44px; margin-bottom: 30px; color: var(--pixel-cyan); font-size: 0.9em; }
.article-title-grid { grid-template-columns: minmax(0,1.4fr) minmax(240px,.6fr); gap: 52px; }
.article-title-grid h1 { font-size: clamp(2.3rem,6vw,5rem); }
.article-summary { max-width: 67ch; color: #bcc4df; font-family: var(--font-reading); font-size: 1.08rem; }
.article-facts { background: var(--panel); border: 1px solid #354064; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.article-facts > div { grid-template-columns: 1fr; gap: 3px; padding: 14px; border-color: #30395e; }
.article-facts dt { color: var(--pixel-cyan); font-size: .63rem; letter-spacing: .06em; text-transform: uppercase; }
.article-facts dd { color: var(--ink); }
.article-layout { grid-template-columns: minmax(150px,.35fr) minmax(0,var(--reading)) minmax(190px,.45fr); gap: 30px; align-items: start; }
.article-rail { top: calc(var(--header-height) + 24px); padding: 16px; color: var(--muted); background: #090d1c; border: 1px solid #30395e; }
.article-rail a { color: var(--pixel-cyan); }
.article-toc { max-height: calc(100vh - 110px); overflow-y: auto; }
.toc-title { color: var(--ink); font-weight: 800; text-transform: uppercase; }
.article-toc a { display: block; padding: 7px 8px; color: var(--muted); border-left: 2px solid #30395e; }
.article-toc a: hover, .article-toc a: focus-visible { color: var(--pixel-cyan); background: #11162c; border-color: var(--pixel-cyan); }
.article-body { min-width: 0; padding: clamp(22px,4vw,44px); color: #d6dbed; background: #0b0f20; border: 1px solid #354064; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.article-body h2, .article-body h3, .article-body h4 { scroll-margin-top: calc(var(--header-height) + 28px); color: var(--ink); font-family: var(--font-reading); }
.article-body h2 { padding: 14px 0 10px; margin-top: 52px; border-top: 4px solid var(--pixel-cyan); border-bottom: 1px solid #30395e; }
.article-body h2: before { margin-right: 10px; color: var(--pixel-cyan); content: "///"; }
.article-body h3 { margin-top: 38px; color: var(--pixel-gold); }
.article-body a { color: var(--pixel-cyan); text-decoration: underline; text-decoration-thickness: 2px; text-underline-offset: 3px; }
.article-body blockquote { padding: 18px 20px; color: #cbd2e8; background: #121831; border: 0; border-left: 5px solid var(--pixel-gold); }
.article-body pre { overflow-x: auto; padding: 44px 18px 18px; color: #d7fbc4; background: #050711; border: 1px solid #3a4465; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.article-body pre: before { position: absolute; top: 10px; left: 14px; color: var(--pixel-green); content: "> TERMINAL_OUTPUT"; font-size: 0.8em; }
.article-body code { color: #b9bf6d; background: #141a2f; border: 1px solid #2e3858; }
.article-body pre code { background: transparent; border: 0; }
.article-body img { border: 2px solid #465176; box-shadow: 6px 6px 0 var(--pixel-shadow); }
.article-body table { display: block; max-width: 100%; overflow-x: auto; border-color: #354064; }
.article-body th { color: var(--pixel-cyan); background: #11162d; }
.article-body th, .article-body td { border-color: #354064; }
.copy-code { top: 8px; right: 8px; min-width: 72px; min-height: 34px; color: #071018; background: var(--pixel-green); border: 1px solid #354064; }
.article-footer { grid-template-columns: minmax(0,.5fr) minmax(0,1.5fr); gap: 36px; padding-top: 62px; }
.article-pagination a { padding: 18px; background: var(--panel); border: 1px solid #354064; box-shadow: 4px 4px 0 var(--pixel-shadow); }
.article-pagination a: hover, .article-pagination a: focus-visible { color: var(--pixel-cyan); border-color: #6875a8; }

.contact-page { min-height: auto; padding-top: 80px; padding-bottom: 96px; }
.contact-lead { margin-bottom: 50px; }
.contact-lead > p: last-child { max-width: 65ch; font-family: var(--font-reading); font-size: 1.08rem; }
.contact-grid { grid-template-columns: minmax(0,1.25fr) minmax(260px,.75fr); gap: 28px; }
.contact-form, .contact-aside { padding: clamp(22px,4vw,36px); background: var(--panel); border: 2px solid #465176; box-shadow: 7px 7px 0 var(--pixel-shadow); }
.contact-form label > span { color: var(--pixel-cyan); font-size: .68rem; font-weight: 800; letter-spacing: .06em; text-transform: uppercase; }
.contact-form input, .contact-form textarea { color: var(--ink); background: #070a18; border: 2px solid #354064; }
.contact-form input: focus, .contact-form textarea: focus { border-color: var(--pixel-cyan); box-shadow: 0 0 3px #1f3150; }
.contact-aside { align-content: start; gap: 28px; }
.contact-email { color: var(--pixel-cyan); font-size: clamp(.8rem,1.6vw,1rem); }
.contact-socials a { min-height: 44px; padding: 10px; background: #11162c; border: 1px solid #30395e; }
.contact-socials a: hover, .contact-socials a: focus-visible { color: #071018; background: var(--pixel-cyan); }
.availability-note { padding: 14px; color: #cbd2e8; background: #11162c; border: 1px solid #354064; }

```

```
.site-footer { position: relative; padding-top: 70px; color: var(--ink); background: #070a18; border-top: 2px solid #465176; }
.footer-grid { gap: 48px; }
.footer-grid h2 { color: var(--ink); font-size: clamp(1.8rem,4vw,3.4rem); text-transform: uppercase; text-shadow: 3px 3px 0 #202746; }
.footer-contact { padding: 20px; background: #0c1023; border: 1px solid #354064; box-shadow: 5px 5px 0 var(--pixel-shadow); }
.footer-links a { display: inline-flex; align-items: center; min-height: 44px; color: var(--muted); }
.footer-links a:hover, .footer-links a:focus-visible { color: var(--pixel-cyan); }
.footer-bottom { padding-block: 20px; margin-top: 60px; color: var(--muted); border-top: 1px solid #30395e; font-size: .66rem; letter-sp
.footer-bottom a { min-height: 44px; color: var(--pixel-cyan); }
.not-found { min-height: 70vh; padding-block: 100px; }
:focus-visible { outline: 3px solid var(--pixel-cyan); outline-offset: 4px; }

html.motion-enabled main { opacity: .99; transform: translateY(4px); }
html.motion-ready [data-motion] { transition-duration: 260ms,300ms,300ms; transition-timing-function: ease,steps(4,end),steps(4,end); tr
html.motion-ready [data-motion="grid"]:not([data-motion-state="visible"]) { clip-path: inset(0 0 10% 0); transform: translateY(12px); }
html.motion-ready [data-motion="hero"]:not([data-motion-state="visible"]), html.motion-ready [data-motion="reveal"]:not([data-motion-sta

@keyframes world-boot { from { opacity: 0; clip-path: inset(0 100% 0 0); } to { opacity: 1; clip-path: inset(0 0 0 0); } }
@keyframes circuit-load { to { stroke-dashoffset: 0; } }
@keyframes bot-spawn { to { opacity: 1; transform: translateY(0); } }
@keyframes pixel-blink { 0%,100% { opacity: .45; } 50% { opacity: 1; } }
@media (max-width: 1080px) {
  .site-nav a { padding-inline: 9px; font-size: .63rem; }
  .hero-grid { grid-template-columns: minmax(0,1fr) minmax(220px,290px); width: calc(100% - 34px); }
  .expertise-grid { grid-template-columns: repeat(2,minmax(0,1fr)); }
  .article-layout { grid-template-columns: minmax(0,1fr) minmax(190px,.32fr); }
  .article-rail-left { display: none; }
}

@media (max-width: 820px) {
  :root { --header-height: 64px; }
  .shell { width: min(calc(100% - 28px), var(--shell)); }
  .section { padding-block: 64px; }
  .menu-toggle { display: inline-flex; gap: 12px; align-items: center; padding: 8px 11px; font-size: .68rem; font-weight: 800; text-tran
  .site-nav { position: absolute; top: calc(100% + 2px); right: 14px; left: 14px; display: none; padding: 10px; background: #080b1a; bor
  .site-nav.is-open { display: grid; }
  .site-nav a { width: 100%; min-height: 48px; padding-inline: 14px; font-size: .7rem; }
  .hero-shell { display: flex; flex-direction: column; }
  .pixel-world { order: 2; min-height: 270px; }
  .pixel-world svg { min-width: 680px; min-height: 270px; transform: translateX(-80px); }
  .pixel-world-label { top: 12px; right: 12px; font-size: .55rem; }
  .hero-grid { grid-template-columns: 1fr; order: 1; gap: 18px; width: calc(100% - 20px); margin: 0 auto 18px; }
  .hero-portrait { order: -1; justify-self: center; width: min(100%,280px); }
  ... [Truncated 861 lines in PDF preview - see TXT bundle for full text]
}
```

File (8/30): app\globals.css

```
:root {
  color-scheme: dark;
  --paper: #060814;
  --paper-deep: #0b0f21;
  --ink: #eef1ff;
  --ink-soft: #c6cdea;
  --muted: #9ca7ca;
  --line: #30395e;
  --signal: #55d5d8;
  --signal-dark: #55d5d8;
  --night: #050713;
  --night-soft: #10152c;
  --night-line: #30395e;
  --night-text: #eef1ff;
  --white: #0c1023;
  --shell: 1180px;
  --reading: 742px;
  --radius: 0;
  --header-height: 68px;
  --font-ui: "Cascadia Code", "SFMono-Regular", Consolas, monospace;
  --font-editorial: "Cascadia Code", "SFMono-Regular", Consolas, monospace;
  --font-mono: "Cascadia Code", "SFMono-Regular", Consolas, monospace;
}

* {
  box-sizing: border-box;
}

html {
```

```
    overflow-x: hidden;
    color-scheme: dark;
    background: var(--night);
    scroll-behavior: smooth;
    scroll-padding-top: calc(var(--header-height) + 28px);
}
```

```
body {
    margin: 0;
    min-width: 320px;
    overflow-x: hidden;
    color: var(--ink);
    background: var(--night);
    font-family: var(--font-ui);
    font-size: 15px;
    line-height: 1.65;
    text-rendering: optimizeLegibility;
    -webkit-font-smoothing: antialiased;
}
```

```
body::before {
    position: fixed;
    inset: 0;
    z-index: -1;
    pointer-events: none;
    content: "";
    border: 14px solid rgba(21, 23, 19, 0.025);
}
```

```
::selection {
    color: var(--paper);
    background: var(--signal);
}
```

```
img {
    display: block;
    max-width: 100%;
}
```

```
a {
    color: inherit;
    text-decoration: none;
}
```

```
button,
input,
textarea {
    font: inherit;
}
```

```
button,
a {
    -webkit-tap-highlight-color: transparent;
}
```

```
h1,
h2,
h3,
p {
    margin-top: 0;
}
```

```
h1,
h2 {
    font-family: var(--font-editorial);
    font-weight: 500;
    letter-spacing: -0.045em;
}
```

```
h3 {
    letter-spacing: -0.025em;
}
```

```
.shell {
    width: min(calc(100% - 48px), var(--shell));
    margin-inline: auto;
}
```

```
.section {
  padding-block: 96px;
}

.page-section {
  padding-block: 32px 104px;
}

.skip-link {
  position: fixed;
  top: 10px;
  left: 10px;
  z-index: 100;
  padding: 10px 14px;
  color: var(--paper);
  background: var(--ink);
  transform: translateY(-160%);
}

.skip-link:focus {
  transform: translateY(0);
}

.sr-only {
  position: absolute;
  width: 1px;
  height: 1px;
  padding: 0;
  margin: -1px;
  overflow: hidden;
  clip: rect(0, 0, 0, 0);
  white-space: nowrap;
  border: 0;
}

.mono {
  font-family: var(--font-mono);
  font-size: 0.77rem;
  letter-spacing: 0.04em;
  text-transform: uppercase;
}

.muted {
  color: var(--muted);
}

.eyebrow {
  margin-bottom: 16px;
  color: var(--signal-dark);
  font-family: var(--font-mono);
  font-size: 0.76rem;
  font-weight: 700;
  letter-spacing: 0.11em;
  text-transform: uppercase;
}

.site-header {
  position: sticky;
  top: 0;
  z-index: 50;
  height: var(--header-height);
  background: rgba(242, 240, 233, 0.94);
  border-bottom: 1px solid var(--line);
  backdrop-filter: blur(14px);
}

.header-inner {
  display: flex;
  align-items: center;
  justify-content: space-between;
  height: 100%;
}

.brand {
  display: inline-flex;
  gap: 12px;
  align-items: center;
  font-size: 0.93rem;
```

```

    font-weight: 700;
    letter-spacing: -0.02em;
}

.brand-mark {
    display: grid;
    width: 34px;
    height: 34px;
    place-items: center;
    color: var(--paper);
    background: var(--ink);
    border-radius: 50%;
    font-family: var(--font-mono);
    font-size: 0.78rem;
}

.site-nav {
    display: flex;
    gap: 30px;
    align-items: center;
}

.site-nav a {
    position: relative;
    color: var(--muted);
    font-size: 0.85rem;
    font-weight: 650;
}

.site-nav a::after {
    position: absolute;
    right: 0;
    bottom: -8px;
    left: 0;
    height: 2px;
    content: "";
    background: var(--signal);
    transform: scaleX(0);
    transform-origin: right;
    transition: transform 180ms ease;
}

.site-nav a:hover,
.site-nav a.active {
    color: var(--ink);
}

.site-nav a:hover::after,
.site-nav a.active::after {
    transform: scaleX(1);
    transform-origin: left;
}

.menu-toggle {
    display: none;
    gap: 10px;
    align-items: center;
    padding: 8px 0;
    color: var(--ink);
    background: transparent;
    border: 0;
    cursor: pointer;
    font-size: 0.84rem;
    font-weight: 700;
}

.hero {
    padding-top: 80px;
    border-bottom: 1px solid var(--line);
}

.hero-grid {
    display: grid;
    grid-template-columns: minmax(0, 1.36fr) minmax(290px, 0.64fr);
    gap: 88px;
    align-items: center;
    min-height: 620px;
    padding-bottom: 74px;
}

```

```

}

.hero-copy {
  padding-top: 20px;
}

.hero-kicker {
  display: flex;
  gap: 10px;
  align-items: center;
  margin-bottom: 30px;
  color: var(--muted);
  font-family: var(--font-mono);
  font-size: 0.75rem;
  letter-spacing: 0.045em;
  text-transform: uppercase;
}

.status-dot {
  display: inline-block;
  flex: 0 0 auto;
  width: 8px;
  height: 8px;
  background: #5a8f42;
  border-radius: 50%;
  box-shadow: 0 0 0 4px rgba(90, 143, 66, 0.13);
}

.hero h1 {
  max-width: 870px;
  margin-bottom: 28px;
  font-size: clamp(3.7rem, 6.8vw, 5.4rem);
  line-height: 0.95;
}

.hero-lede {
  max-width: 630px;
  ... [Truncated 1951 lines in PDF preview - see TXT bundle for full text]

```

File (9/30): applicon.svg

```

<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 64 64">
  <rect width="64" height="64" fill="#070a18"/>
  <rect x="5" y="5" width="54" height="54" fill="none" stroke="#55d8df" stroke-width="4"/>
  <path d="M16 46 27 17h8l11 29h-8l-2-7H26l-2 7h-8Zm12-14h6l-3-10-3 10Z" fill="#dce2f8"/>
  <path d="M46 17h5v29h-5z" fill="#8dea78"/>
</svg>

```

File (10/30): app\layout.js

```

import './globals.css';
import './game-theme.css';
import './series-theme.css';
import { JsonLd } from '@components/JsonLd';
import { MotionSystem } from '@components/MotionSystem';
import { SiteFooter } from '@components/SiteFooter';
import { SiteHeader } from '@components/SiteHeader';
import { profile } from '@data/portfolio';
import { createSiteJsonLd, socialImage, twitterImage } from '@lib/seo';
import { absoluteUrl, siteConfig } from '@lib/site';

export const metadata = {
  metadataBase: new URL(siteConfig.url),
  title: {
    default: siteConfig.title,
    template: `%s | ${siteConfig.name}`
  },
  description: siteConfig.description,
  alternates: {
    canonical: absoluteUrl("/")
  },
  authors: [

```

```

    {
      name: profile.name,
      url: absoluteUrl("/about/")
    }
  ],
  creator: profile.name,
  category: "technology",
  openGraph: {
    type: "website",
    locale: "en_US",
    title: siteConfig.title,
    description: siteConfig.description,
    url: absoluteUrl("/"),
    siteName: siteConfig.name,
    images: [
      {
        url: socialImage,
        width: 1200,
        height: 630,
        alt: `${profile.name}, ${profile.role}`
      }
    ]
  },
  twitter: {
    card: "summary_large_image",
    title: siteConfig.title,
    description: siteConfig.description,
    images: [twitterImage]
  },
  robots: {
    index: true,
    follow: true,
    googleBot: {
      index: true,
      follow: true,
      "max-image-preview": "large",
      "max-snippet": -1,
      "max-video-preview": -1
    }
  }
}
};

export default function RootLayout({ children }) {
  return (
    <html lang="en" data-scroll-behavior="smooth">
      <body>
        <JsonLd data={createSiteJsonLd()} />
        <MotionSystem />
        <a className="skip-link" href="#main-content">
          Skip to content
        </a>
        <SiteHeader />
        <main id="main-content">{children}</main>
        <SiteFooter />
      </body>
    </html>
  );
}

```

File (11/30): app\not-found.js

```

import Link from "next/link";

export default function NotFound() {
  return (
    <section className="not-found shell">
      <p className="eyebrow">404 / Page not found</p>
      <h1>This page could not be found.</h1>
      <p>The address may have changed, or the article may still be a draft.</p>
      <Link className="button primary" href="/">
        Return to the portfolio <span aria-hidden="true">&rarr</span>
      </Link>
    </section>
  );
}

```

File (12/30): app\page.js

```
import Link from "next/link";
import { ProjectCard } from "@components/ProjectCard";
import { PostCard } from "@components/PostCard";
import { SectionHeading } from "@components/SectionHeading";
import { PixelWorld } from "@components/PixelWorld";
import {
  education,
  experience,
  profile,
  projects,
  publication,
  technologyGroups,
  toolkitHeading,
  toolkitIntro,
  workingMethod
} from "@data/portfolio";
import { getAllPosts } from "@lib/content";

export default function HomePage() {
  const posts = getAllPosts();
  const featuredProjects = projects.filter((project) => project.featured).slice(0, 4);
  const competitionProject = projects.find(
    (project) => project.slug === "megasumo-autonomous-robot"
  );

  return (
    <>
    < /* 1. Hero: client outcome + primary project CTA */
    <section className="hero" data-mission="origin" data-mission-label="ORIGIN" data-mission-num="01">
      <div className="shell hero-shell">
        <PixelWorld />
        <div className="hero-grid">
          <div className="hero-copy">
            <div className="hero-kicker">
              <span className="status-dot" aria-hidden="true" />
              {profile.label}
            </div>
            <h1 className="identity-title">
              <span className="identity-name">{profile.name}</span>
              <span className="identity-role">{profile.role}</span>
            </h1>
            <p className="hero-lede">
              {profile.headline}
            </p>
            <p className="hero-sub">
              {profile.summary}
            </p>
            <div className="hero-actions">
              <Link className="button primary" href="/contact">
                Tell me about your project <span aria-hidden="true">&arr</span>
              </Link>
              <Link className="button text-button" href="#method">
                See how I solve problems <span aria-hidden="true">&darr</span>
              </Link>
            </div>
          </div>
          <figure className="hero-portrait">
            <img src={profile.portrait} alt={`Portrait of ${profile.name}`} />
          </figure>
        </div>
      </div>
    </section>

    < /* 2. Credibility row */
    <section className="section home-about-section" data-mission="credentials" data-mission-label="CREDENTIALS" data-mission-num="02">
      <div className="shell home-about-grid">
        <div className="home-about-copy">
          <p className="eyebrow">Background</p>
          <h2>From problem to working prototype, across the full system.</h2>
          <p>
            I trace a problem from sensor or board through transport, application state,
            and the operator interface. Teaching shapes how I engineer: I document
            assumptions, explain failure modes, and check whether someone else can
          </p>
        </div>
      </div>
    </section>
  )
}
```

```

        understand and use the result.
    </p>
    <Link className="text-link" href="/about">
        Read my full background <span aria-hidden="true">&rarr;</span>
    </Link>
</div>
<dl className="home-about-facts">
    <div>
        <dt>Current role</dt>
        <dd>
            {experience[0].role}
            <span>{experience[0].organization}</span>
        </dd>
    </div>
    {education[0] ? (
        <div>
            <dt>Study</dt>
            <dd>{education[0].credential}</dd>
        </div>
    ) : null}
    <div>
        <dt>Based in</dt>
        <dd>{profile.location}</dd>
    </div>
    <div>
        <dt>Core focus</dt>
        <dd>Hardware prototypes, IoT products, robotics, and engineering education</dd>
    </div>
</dl>
</div>
</section>

{/* 3. Featured projects */}
<section className="section shell" data-mission="projects" data-mission-label="PROJECTS" data-mission-num="03">
    <SectionHeading
        eyebrow="Selected work"
        title="Projects built around a problem, a build, and a result."
        action={
            <Link className="text-link" href="/work">
                View all projects <span aria-hidden="true">&rarr;</span>
            </Link>
        }
    />
    <div className="project-grid">
        {featuredProjects.map((project, index) => (
            <ProjectCard key={project.slug} project={project} index={index} />
        ))}
    </div>
</section>

{/* Experience and results strip */}
<section className="section home-proof-section">
    <div className="shell">
        <SectionHeading
            eyebrow="Experience and results"
            title="Teaching, research, and competition work."
        />
        <div className="proof-strip">
            <article className="proof-card">
                <span className="proof-index mono">01</span>
                <p>
                    <h3>University teaching</h3>
                    <p>{experience[0].role}</p>
                    <p>{experience[0].organization}</p>
                    <Link className="text-link" href="/about#experience">
                        View experience <span aria-hidden="true">&rarr;</span>
                    </Link>
                </p>
            </article>
            <article className="proof-card">
                <span className="proof-index mono">02</span>
                <p>
                    <h3>Research publication</h3>
                    <p>{publication.title}</p>
                    <p>{publication.description}</p>
                    <Link className="text-link" href="/about#publications">
                        View publication record <span aria-hidden="true">&rarr;</span>
                    </Link>
                </p>
            </article>
            <article className="proof-card">
                <span className="proof-index mono">03</span>

```

```

        <p className="proof-label">Robotics competition</p>
        <h3>{competitionProject.title}</h3>
        <p>{competitionProject.outcome}</p>
        <Link className="text-link" href="{`/work#${competitionProject.slug}`}">
            View project <span aria-hidden="true">&rarr;</span>
        </Link>
    </article>
</div>
</div>
</section>

/* 4. Working method */
<section id="method" className="section section-ink home-method-section" data-mission="method" data-mission-label="METHOD" data-
    <div className="shell">
        <SectionHeading
            eyebrow="How I work"
            title="Question, learn, build, test."
        </>
        <div className="method-signal-path" aria-hidden="true">
            <svg viewBox="0 0 1000 12" preserveAspectRatio="none" className="method-svg-path">
                <line x1="0" y1="6" x2="1000" y2="6" className="method-path-track" />
                <line x1="0" y1="6" x2="1000" y2="6" className="method-path-pulse" />
            </svg>
        </div>
        <div className="method-grid">
            {workingMethod.map((item) => (
                <article key={item.step} className="method-step" data-step={item.step}>
                    <span className="method-step-index mono">{item.step}</span>
                    <h3>{item.label}</h3>
                    <p>{item.description}</p>
                </article>
            ))}
        </div>
    </div>
</section>

/* 5. Toolkit */
<section className="section shell home-toolkit-section" data-mission="toolkit" data-mission-label="TOOLKIT" data-mission-num="0
    <SectionHeading
        eyebrow="Toolkit"
        title={toolkitHeading}
        action={
            <Link className="text-link" href="/about#skills">
                View full toolkit <span aria-hidden="true">&rarr;</span>
            </Link>
        }
    </>
    <p className="section-intro toolkit-intro">{toolkitIntro}</p>
    <div className="technology-matrix">
        {technologyGroups.map((group) => (
            <article className="technology-cluster" key={group.index} data-node={group.index}>
                <div className="technology-cluster-top">
                    <span className="technology-cluster-index mono">{group.index}</span>
                    <span className="mono">SYSTEM CLASS</span>
                </div>
                <h3>{group.title}</h3>
                <p>{group.description}</p>
                <ul className="technology-tool-list" aria-label={` ${group.title} tools`>
                    {group.tools.map((tool, i) => (
                        <li key={tool}>
                            <span className="mono" aria-hidden="true">
                                {String(i + 1).padStart(2, "0")}
                            </span>
                            <strong>{tool}</strong>
                        </li>
                    ))}
                </ul>
            </article>
        ))}
    </div>
</section>

/* 6. Curiosity origin teaser */
<section className="section home-origin-section" data-mission="story" data-mission-label="STORY" data-mission-num="06">
    <div className="shell home-origin-grid">
        <div className="home-origin-copy">
            <p className="eyebrow">Where it started</p>
            <h2>Titles tell you where someone works. Questions tell you how they think.</h2>

```

```

    <p>
      Mine started with an ATM. I wanted to know how it recognized an account,
      checked whether money was available, and exchanged information with systems
      somewhere else. That curiosity moved into Windows CMD, BIOS passwords, online
      games, electronics, networks, and eventually the machines and connected
      products I build today.
    </p>
    <Link className="text-link" href="/about">
      Read the full story <span aria-hidden="true">&rarr;</span>
    </Link>
  </div>
</div>
</section>

/* 7. Writing and teaching */
<section className="section shell" data-mission="writing" data-mission-label="WRITING" data-mission-num="07">
  <SectionHeading
    eyebrow="Writing and tutorials"
    title="Linux, embedded systems, Flutter, and security walkthroughs."
    action={
      <Link className="text-link" href="/writing">
        Browse all writing <span aria-hidden="true">&rarr;</span>
      </Link>
    }
  />
  <div className="post-list">
    {posts.slice(0, 3).map((post, index) => (
      <PostCard key={post.slug} post={post} index={index} />
    ))}
  </div>
</section>

/* 8. Structured project-brief CTA */
<section className="section home-contact-section" data-mission="contact" data-mission-label="CONTACT" data-mission-num="08">
  <div className="shell home-contact-grid">
    <div>
      <p className="eyebrow">Contact</p>
      <h2>Bring me the problem, even if the solution is not clear yet.</h2>
      <p>
        Tell me what you are trying to build or fix, what already exists, and where
        you are stuck. I will tell you whether I can help and what the next useful
        step should be.
      </p>
    </div>
    <div className="home-contact-actions">
      <Link className="button primary" href="/contact">
        Send your project brief <span aria-hidden="true">&rarr;</span>
      </Link>
      <a className="button text-button" href={`mailto:${profile.email}`}>
        Email directly <span aria-hidden="true">&rarr;</span>
      </a>
    </div>
  </div>
</section>
</>
);
}

```

File (13/30): app\robots.js

```

export const dynamic = "force-static";

import { siteConfig } from "@lib/site";

export default function robots() {
  return {
    rules: {
      userAgent: "*",
      allow: "/"
    },
    sitemap: `${siteConfig.url}/sitemap.xml`
  };
}

```

File (14/30): app\series-theme.css

```
/* Writing-series routes, publication classifications, and social brand marks. */
.writing-series-section {
  padding-top: 72px;
  padding-bottom: 20px;
}

.writing-series-heading,
.writing-library-heading {
  grid-template-columns: minmax(0, 1fr) auto;
  align-items: end;
}

.series-count {
  color: var(--pixel-cyan);
  font-size: 0.64rem;
  letter-spacing: 0.06em;
}

.writing-series-grid {
  display: grid;
  grid-template-columns: minmax(0, 0.9fr) minmax(0, 1.1fr);
  gap: 18px;
}

.writing-series-panel {
  position: relative;
  min-width: 0;
  padding: 22px;
  background: #0c1023;
  border: 2px solid #3b456a;
  box-shadow: 6px 6px 0 var(--pixel-shadow);
}

.writing-series-panel::before {
  position: absolute;
  top: -2px;
  left: 22px;
  width: 64px;
  height: 4px;
  content: "";
  background: var(--pixel-cyan);
}

.learn-series::before {
  background: var(--pixel-green);
}

.series-panel-top {
  display: flex;
  justify-content: space-between;
  gap: 18px;
  color: var(--muted);
  font-size: 0.6rem;
  letter-spacing: 0.06em;
}

.series-panel-top span:first-child {
  color: var(--pixel-gold);
}

.series-panel-copy {
  padding: 48px 0 28px;
}

.series-label {
  margin: 0 0 8px;
  color: var(--pixel-cyan);
  font-size: 0.68rem;
  font-weight: 900;
  letter-spacing: 0.08em;
  text-transform: uppercase;
}

.series-panel-copy h3 {
  max-width: 18ch;
  margin: 0 0 14px;
```

```

    font-size: clamp(1.55rem, 3vw, 2.5rem);
    text-transform: uppercase;
}

.series-panel-copy > p:last-child {
    max-width: 62ch;
    margin: 0;
    color: #adb6d5;
    font-family: var(--font-reading);
    line-height: 1.7;
}

.series-entry-list {
    display: grid;
    gap: 8px;
    padding: 0;
    margin: 0;
    list-style: none;
}

.series-entry {
    display: grid;
    grid-template-columns: auto minmax(0, 1fr);
    gap: 12px;
    align-items: center;
    min-height: 50px;
    padding: 9px 10px;
    background: #080b1a;
    border: 1px solid #30395e;
}

.series-entry-state {
    padding: 5px 7px;
    color: #071018;
    background: var(--pixel-cyan);
    font-family: var(--font-mono);
    font-size: 0.56rem;
    font-weight: 900;
    letter-spacing: 0.04em;
    text-transform: uppercase;
}

.series-entry-state.draft {
    background: var(--pixel-gold);
}

.series-entry-title {
    min-width: 0;
    overflow-wrap: anywhere;
    color: #dce2f8;
    font-size: 0.78rem;
    line-height: 1.4;
}

a.series-entry-title:hover,
a.series-entry-title:focus-visible {
    color: var(--pixel-cyan);
}

.series-empty {
    padding: 16px;
    color: #939dbe;
    border: 1px dashed #465176;
    font-family: var(--font-reading);
}

.learn-track-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 10px;
}

.learn-track {
    min-width: 0;
    padding: 16px;
    background: #080b1a;
    border: 1px solid #354064;
}

```

```

.learn-track-top {
  display: flex;
  justify-content: space-between;
  gap: 10px;
  align-items: center;
  color: var(--pixel-cyan);
  font-size: 0.58rem;
}

.learn-track-status {
  color: var(--pixel-green);
  text-align: right;
  text-transform: uppercase;
}

.learn-track h4 {
  margin: 26px 0 10px;
  color: #dce2f8;
  font-size: 1rem;
  text-transform: uppercase;
}

.learn-track &gt; p {
  margin: 0;
  color: #9fa9ca;
  font-family: var(--font-reading);
  font-size: 0.85rem;
  line-height: 1.6;
}

.learn-track .learn-track-empty {
  padding-top: 16px;
  margin-top: 18px;
  color: #7f8aad;
  border-top: 1px solid #30395e;
  font-size: 0.72rem;
}

.series-entry-list.compact {
  margin-top: 18px;
}

.series-entry-list.compact .series-entry {
  grid-template-columns: 1fr;
}

.writing-library-heading {
  margin-bottom: 30px;
}

.publication-classification {
  display: flex;
  flex-wrap: wrap;
  gap: 8px;
  margin-bottom: 14px;
}

.publication-classification span {
  display: inline-flex;
  align-items: center;
  min-height: 28px;
  padding: 5px 9px;
  font-family: var(--font-mono);
  font-size: 0.58rem;
  font-weight: 900;
  letter-spacing: 0.05em;
  text-transform: uppercase;
}

.publication-ranking {
  color: #071018;
  background: var(--pixel-green);
  border: 1px solid #b3f89b;
}

.publication-type {
  color: var(--pixel-cyan);
  background: #10162d;
}

```

```

    border: 1px solid #3f7290;
}

.social-link {
  gap: 8px;
}

.social-icon {
  display: block;
  width: 18px;
  height: 18px;
  flex: 0 0 18px;
  fill: currentColor;
}

.contact-socials a {
  display: grid;
  grid-template-columns: 20px minmax(0, 1fr) auto;
  gap: 10px;
  align-items: center;
}

.contact-socials .social-link-text {
  min-width: 0;
}

.contact-socials .social-link-arrow {
  color: var(--pixel-cyan);
}

.contact-socials a:hover .social-link-arrow,
.contact-socials a:focus-visible .social-link-arrow {
  color: #071018;
}

@media (max-width: 820px) {
  .writing-series-heading,
  .writing-library-heading,
  .writing-series-grid {
    grid-template-columns: 1fr;
  }

  .series-count {
    justify-self: start;
  }
}

@media (max-width: 560px) {
  .writing-series-section {
    padding-top: 52px;
  }

  .writing-series-panel {
    padding: 18px;
    box-shadow: 4px 4px 0 var(--pixel-shadow);
  }

  .series-panel-top {
    display: grid;
  }

  .series-panel-copy {
    padding-top: 36px;
  }

  .learn-track-grid {
    grid-template-columns: 1fr;
  }
... [Truncated 17 lines in PDF preview - see TXT bundle for full text]

```

File (15/30): app\sitemap.js

```

export const dynamic = "force-static";

import { getAllPosts } from "@lib/content";
import { siteConfig } from "@lib/site";

```

```

const routes = [
  { pathname: "", changeFrequency: "weekly", priority: 1 },
  { pathname: "/work", changeFrequency: "monthly", priority: 0.9 },
  { pathname: "/about", changeFrequency: "monthly", priority: 0.9 },
  { pathname: "/writing", changeFrequency: "weekly", priority: 0.9 },
  { pathname: "/contact", changeFrequency: "yearly", priority: 0.6 }
];

export default function sitemap() {
  const pages = routes.map((route) => ({
    url: `${siteConfig.url}${route.pathname}/`,
    lastModified: new Date("2026-07-25"),
    changeFrequency: route.changeFrequency,
    priority: route.priority
  }));

  const posts = getAllPosts().map((post) => ({
    url: `${siteConfig.url}/writing/${post.slug}/`,
    lastModified: new Date(post.updatedAt || post.publishedAt),
    changeFrequency: "monthly",
    priority: 0.7
  }));

  return [...pages, ...posts];
}

```

File (16/30): app\work\page.js

```

import { ProjectCard } from "@components/ProjectCard";
import { projects } from "@data/portfolio";
import { createPageMetadata } from "@lib/seo";

export const metadata = createPageMetadata({
  title: "Embedded Systems and Robotics Projects",
  description:
    "Explore embedded systems and robotics projects by Ahmed Asl, including ESP32 firmware, IoT telemetry, secure hardware, OTA tools, and more.",
  pathname: "/work/"
});

export default function WorkPage() {
  return (
    <div>
      <section className="page-intro shell">
        <p className="eyebrow">Selected engineering work</p>
        <h1>Embedded Systems, IoT, and Robotics Projects</h1>
        <p className="page-lede">
          Explore my work in ESP32 firmware, IoT telemetry, autonomous robots,
          secure access systems, OTA management, and Flutter control interfaces.
        </p>
      </section>

      <section className="shell page-section" aria-labelledby="project-archive-title">
        <div className="archive-header">
          <h2>Project archive</h2>
          <span className="mono muted">{projects.length} projects</span>
          <span className="mono muted">Hardware, firmware, and interface work</span>
        </div>
        <div className="project-grid">
          {projects.map((project, index) => (
            <ProjectCard
              key={project.slug}
              project={project}
              index={index}
              compact
            />
          ))}
        </div>
      </section>
    </div>
  );
}

```

File (17/30): app\writing\[slug]\page.js

```
import Link from "next/link";
import { notFound } from "next/navigation";
import { ArticleEnhancer } from "@components/ArticleEnhancer";
import { JsonLd } from "@components/JsonLd";
import {
  formatDate,
  getAllPosts,
  getPost,
  renderPost
} from "@lib/content";
import { createArticleJsonLd, createPageMetadata } from "@lib/seo";

export function generateStaticParams() {
  return getAllPosts().map((post) => ({ slug: post.slug }));
}

export const dynamicParams = false;

export async function generateMetadata({ params }) {
  const { slug } = await params;
  const post = getPost(slug);
  if (!post) return {};

  const metadata = createPageMetadata({
    title: post.title,
    description: post.summary,
    pathname: `/writing/${post.slug}/`
  });

  return {
    ...metadata,
    openGraph: {
      ...metadata.openGraph,
      type: "article",
      publishedTime: post.publishedAt,
      modifiedTime: post.updatedAt || post.publishedAt,
      tags: post.tags
    }
  };
}

export default async function ArticlePage({ params }) {
  const { slug } = await params;
  const post = getPost(slug);
  if (!post || post.draft) notFound();

  const { html, headings } = renderPost(post.content);
  const posts = getAllPosts();
  const currentIndex = posts.findIndex((item) => item.slug === post.slug);
  const previous = posts[currentIndex + 1];
  const next = posts[currentIndex - 1];
  const wasUpdated = post.updatedAt & post.updatedAt !== post.publishedAt;

  return (
    <article className="article-page">
      <JsonLd data={createArticleJsonLd(post)} />

      <header className="article-header shell">
        <Link className="article-back" href="/writing">
          <span aria-hidden="true">&larr;</span> All engineering notes
        </Link>
      </div>
      <div className="article-title-grid">
        <div>
          <p className="eyebrow">
            {post.category}
            {post.series ? ` / ${post.series}` : ""}
          </p>
          <h1>{post.title}</h1>
          <p className="article-summary">{post.summary}</p>
        </div>
        <dl className="article-facts">
          <div>
            <dt>Published</dt>
            <dd>{formatDate(post.publishedAt)}</dd>
          </div>
          {wasUpdated ? (
```

```

        <div>
            <dt>Updated</dt>
            <dd>{{formatDate(post.updatedAt)}}</dd>
        </div>
    ) : null
</div>
    <dt>Reading time</dt>
    <dd>{{post.readingTime} minutes}</dd>
</div>
    {{post.difficulty ? (
        <div>
            <dt>Difficulty</dt>
            <dd>{{post.difficulty}}</dd>
        </div>
    ) : null}}
    {{post.part ? (
        <div>
            <dt>Series position</dt>
            <dd>Part {{post.part}}</dd>
        </div>
    ) : null}}
</div>
</div>
<div className="tag-row article-tags">
    {{post.tags.map((tag) => (
        <span className="tag" key={tag}>
            {{tag}}
        </span>
    ))}}
</div>
</header>

<div className="article-layout shell">
    <aside className="article-rail article-rail-left">
        <p className="mono muted">
            <Link href="/about">Ahmed Asl</Link>
        </p>
        <p>Embedded systems engineer and teaching assistant in Egypt.</p>
        <a href="#article-end">Jump to article end <span aria-hidden="true">&darr;</span></a>
    </aside>

    <div
        className="article-body"
        dangerouslySetInnerHTML={{ __html: html }}
    >

    <aside className="article-rail article-toc">
        <p className="toc-title">On this page</p>
        <ol>
            {{headings.map((heading) => (
                <li className={`depth-${heading.depth}`} key={heading.id}>
                    <a href={`#${heading.id}`}>{{heading.text}}</a>
                </li>
            ))}}
        </ol>
    </aside>
</div>

<div id="article-end" className="article-footer shell">
    <div>
        <p className="eyebrow">Read another article</p>
        <h2>Continue with another field note.</h2>
    </div>
    <div className="article-pagination">
        {{previous ? (
            <Link href={`/writing/${previous.slug}`}>
                <span>Previous</span>
                <strong>{{previous.title}}</strong>
            </Link>
        ) : null}}
        {{next ? (
            <Link href={`/writing/${next.slug}`}>
                <span>Next</span>
                <strong>{{next.title}}</strong>
            </Link>
        ) : null}}
    </div>
</div>
</div>

```

```

    <ArticleEnhancer />;
  </article>;
});
}

```

File (18/30): app\writing\page.js

```

import { WritingIndex } from "@components/WritingIndex";
import { WritingSeries } from "@components/WritingSeries";
import { WorldGallery } from "@components/WorldGallery";
import { getAllPosts } from "@lib/content";
import { createPageMetadata } from "@lib/seo";

export const metadata = createPageMetadata({
  title: "Engineering Tutorials and Linux Walkthroughs",
  description:
    "Read Ahmed Asl's tutorials on Linux, cybersecurity, embedded systems, Python, Git, Dart, Flutter, IoT, and machine learning.",
  pathname: "/writing/"
});

export default function WritingPage() {
  const allPosts = getAllPosts({ includeDrafts: true });
  const posts = allPosts.filter((post) => !post.draft);

  return (
    <div>
      <section className="page-intro shell writing-intro">
        <p className="eyebrow">Engineering tutorials and field notes</p>
        <h1>Linux Walkthroughs and Engineering Tutorials</h1>
        <p className="page-lede">
          I document the commands, design decisions, tests, and failure modes
          behind embedded systems, Linux, cybersecurity, Flutter, and AI work.
        </p>
      </section>

      <WorldGallery />

      <WritingSeries posts={allPosts} />

      <section
        className="shell page-section writing-index-section"
        id="published-field-logs"
        aria-labelledby="published-field-logs-title"
      >
        <div className="section-heading writing-library-heading">
          <div>
            <p className="eyebrow">Published library</p>
            <h2 id="published-field-logs-title">Published engineering notes</h2>
            <p className="section-intro">
              Search published articles by subject, tool, or category.
            </p>
          </div>
          <span className="series-count mono">
            {String(posts.length).padStart(2, "0")} PUBLISHED
          </span>
        </div>
        <WritingIndex posts={posts} />
      </section>
    </div>
  );
}

```

File (19/30): components\ArticleEnhancer.js

```

"use client";

import { useEffect } from "react";

export function ArticleEnhancer() {
  useEffect(() => {
    const blocks = document.querySelectorAll(".article-body pre");
    const handlers = [];

```

```

blocks.forEach((block) => {
  if (block.querySelector(".copy-code")) return;
  const button = document.createElement("button");
  button.className = "copy-code";
  button.type = "button";
  button.textContent = "Copy";
  const handler = async () => {
    const code = block.querySelector("code")?.textContent || "";
    await navigator.clipboard.writeText(code);
    button.textContent = "Copied";
    window.setTimeout(() => {
      button.textContent = "Copy";
    }, 1500);
  };
  button.addEventListener("click", handler);
  block.appendChild(button);
  handlers.push([button, handler]);
});

return () => {
  handlers.forEach(([button, handler]) => {
    button.removeEventListener("click", handler);
  });
};
}, []);

return null;
}

```

File (20/30): components\JsonLd.js

```

export function JsonLd({ data }) {
  return (
    <script
      type="application/ld+json"
      dangerouslySetInnerHTML={{
        __html: JSON.stringify(data).replace(/</>/g, "\\u003c")
      }}
    />
  );
}

```

File (21/30): components\MotionSystem.js

```

"use client";

import { usePathname } from "next/navigation";
import { useEffect, useLayoutEffect, useState } from "react";

const MISSIONS = [
  { id: "origin", label: "ORIGIN", num: "01" },
  { id: "credentials", label: "CREDENTIALS", num: "02" },
  { id: "projects", label: "PROJECTS", num: "03" },
  { id: "method", label: "METHOD", num: "04" },
  { id: "toolkit", label: "TOOLKIT", num: "05" },
  { id: "story", label: "STORY", num: "06" },
  { id: "writing", label: "WRITING", num: "07" },
  { id: "contact", label: "CONTACT", num: "08" }
];

const motionGroups = [
  { selector: ".hero-copy & *", kind: "hero", stagger: true },
  { selector: ".hero-profile", kind: "portrait" },
  { selector: ".profile-data & div", kind: "grid", stagger: true },
  { selector: ".technology-rail & span", kind: "rail", stagger: true },
  { selector: ".page-intro & *", kind: "hero", stagger: true },
  { selector: ".section-heading", kind: "line" },
  { selector: ".section-heading & *", kind: "reveal", stagger: true },
  { selector: ".project-card", kind: "grid", stagger: true },
  { selector: ".expertise-item", kind: "grid", stagger: true },
  { selector: ".post-card", kind: "grid", stagger: true },
  { selector: ".publication-grid & *", kind: "reveal", stagger: true },

```

```

{ selector: ".about-statement &gt; p", kind: "reveal", stagger: true },
{ selector: ".about-portrait", kind: "clip" },
{ selector: ".timeline-item", kind: "grid", stagger: true },
{ selector: ".tutorial-card", kind: "grid", stagger: true },
{ selector: ".skills-index &gt; *", kind: "reveal", stagger: true },
{ selector: ".skills-list &gt; span", kind: "grid", stagger: true },
{ selector: ".contact-lead &gt; *", kind: "hero", stagger: true },
{ selector: ".contact-form &gt; *", kind: "grid", stagger: true },
{ selector: ".contact-aside &gt; *", kind: "reveal", stagger: true },
{ selector: ".writing-tools", kind: "reveal" },
{ selector: ".article-title-grid &gt; *", kind: "hero", stagger: true },
{ selector: ".article-facts &gt; div", kind: "grid", stagger: true },
{ selector: ".article-toc", kind: "reveal" },
{ selector: ".article-body &gt; *", kind: "reveal" },
{ selector: ".article-footer &gt; *", kind: "reveal", stagger: true },
{ selector: ".footer-grid &gt; *", kind: "reveal", stagger: true }
];

function prepareMotionNodes() {
  const nodes = new Set();

  for (const group of motionGroups) {
    document.querySelectorAll(group.selector).forEach((node, index) => {
      if (!node.dataset.motion) node.dataset.motion = group.kind;
      if (group.stagger) {
        node.style.setProperty("--motion-order", Math.min(index, 7));
      }
      nodes.add(node);
    });
  }

  return [...nodes];
}

export function MotionSystem() {
  const pathname = usePathname();
  const [activeMission, setActiveMission] = useState("origin");
  const isHomepage = pathname === "/" || pathname === "";

  useEffect(() => {
    const root = document.documentElement;
    const reduceMotion = window.matchMedia(
      "(prefers-reduced-motion: reduce)"
    ).matches;
    const nodes = prepareMotionNodes();
    const main = document.querySelector("main");

    root.classList.add("motion-enabled", "motion-ready");
    main?.classList.remove("route-settled");

    const settleFrame = window.requestAnimationFrame(() => {
      window.requestAnimationFrame(() => { main?.classList.add("route-settled"); });
    });

    if (reduceMotion || !("IntersectionObserver" in window)) {
      nodes.forEach((node) => {
        node.dataset.motionState = "visible";
      });
      document.querySelectorAll("[data-mission]").forEach((sec) => {
        sec.dataset.missionState = "visible";
      });
      return () => window.cancelAnimationFrame(settleFrame);
    }

    // Single Observer for motion nodes and section reveals
    const observer = new IntersectionObserver(
      (entries) => {
        for (const entry of entries) {
          if (!entry.isIntersecting) continue;
          entry.target.dataset.motionState = "visible";
          observer.unobserve(entry.target);
        }
      },
      {
        rootMargin: "0px 0px -7% 0px",
        threshold: 0.08
      }
    );
  });
}

```

```

nodes.forEach((node) => observer.observe(node));

return () => {
  window.cancelAnimationFrame(settleFrame);
  observer.disconnect();
};
}, [pathname]));

// Mission Observer for active mission tracking
useLayoutEffect(() => {
  if (!isHomepage) return undefined;

  const missionElements = Array.from(document.querySelectorAll("[data-mission]"));
  if (missionElements.length === 0) return undefined;

  const reduceMotion = window.matchMedia("(prefers-reduced-motion: reduce)").matches;
  if (reduceMotion || !("IntersectionObserver" in window)) {
    setActiveMission("origin");
    document.documentElement.dataset.activeMission = "origin";
    return undefined;
  }

  const missionObserver = new IntersectionObserver(
    (entries) => {
      let bestEntry = null;
      for (const entry of entries) {
        if (entry.isIntersecting) {
          if (!bestEntry || entry.intersectionRatio > bestEntry.intersectionRatio) {
            bestEntry = entry;
          }
        }
      }
      if (bestEntry && bestEntry.target.dataset.mission) {
        const id = bestEntry.target.dataset.mission;
        setActiveMission(id);
        document.documentElement.dataset.activeMission = id;
      }
    },
    {
      rootMargin: "-15% 0px -40% 0px",
      threshold: [0.1, 0.3, 0.6]
    }
  );

  missionElements.forEach((el) => missionObserver.observe(el));

  return () => {
    missionObserver.disconnect();
  };
}, [pathname, isHomepage]);

// Global scroll progress listener
useEffect(() => {
  let frame = 0;

  const updateProgress = () => {
    frame = 0;
    const scrollable =
      document.documentElement.scrollHeight - window.innerHeight;
    const progress = scrollable > 0 ? window.scrollY / scrollable : 0;
    document.documentElement.style.setProperty(
      "--scroll-progress",
      Math.min(1, Math.max(0, progress))
    );
  };

  const onScroll = () => {
    if (frame) return;
    frame = window.requestAnimationFrame(updateProgress);
  };

  updateProgress();
  window.addEventListener("scroll", onScroll, { passive: true });
  window.addEventListener("resize", onScroll);

  return () => {
    if (frame) window.cancelAnimationFrame(frame);
    window.removeEventListener("scroll", onScroll);
  };
}, []);

```

```

        window.removeEventListener("resize", onScroll);
    };
}, []);

// Bounded pointer parallax for hero portrait
useEffect(() => {
    const hero = document.querySelector(".hero");
    const portrait = document.querySelector(".portrait-frame");
    if (!hero || !portrait) return undefined;

    const reduceMotion = window.matchMedia(
        "(prefers-reduced-motion: reduce)"
    ).matches;
    if (reduceMotion) return undefined;

    const onPointerMove = (event) => {
        const bounds = hero.getBoundingClientRect();
        const x = (event.clientX - bounds.left) / bounds.width - 0.5;
        const y = (event.clientY - bounds.top) / bounds.height - 0.5;
        portrait.style.setProperty("--motion-x", `${x * 8}px`);
        portrait.style.setProperty("--motion-y", `${y * 8}px`);
    };

    const reset = () => {
        portrait.style.setProperty("--motion-x", "0px");
        portrait.style.setProperty("--motion-y", "0px");
    };

    hero.addEventListener("pointermove", onPointerMove);
    hero.addEventListener("pointerleave", reset);

    return () => {
        hero.removeEventListener("pointermove", onPointerMove);
        hero.removeEventListener("pointerleave", reset);
    };
}, [pathname]);

const activeIndex = MISSIONS.findIndex((m) => m.id === activeMission);
const activeMissionObj = MISSIONS[activeIndex &#2264; 0 ? activeIndex : 0];

return (
    &lt;&gt;
    &lt;div className="scroll-progress" aria-hidden="true"&gt;
        &lt;span /&gt;
    &lt;/div&gt;

    {isHomepage &amp;&amp; (
        &lt;&gt;
        {/* Mobile compact readout */}
        &lt;div className="mobile-mission-readout" aria-hidden="true"&gt;
            &lt;span className="readout-tag"&gt;MISSION_{activeMissionObj.num}&lt;/span&gt;
            &lt;span className="readout-label"&gt;{activeMissionObj.label}&lt;/span&gt;
        &lt;/div&gt;

        {/* Desktop compact MissionRail */}
        &lt;nav className="mission-rail" aria-hidden="true"&gt;
            &lt;div className="mission-rail-inner"&gt;
                &lt;div className="mission-rail-header"&gt;
                    &lt;span className="rail-title"&gt;SYS_PATH&lt;/span&gt;
                    &lt;span className="rail-status"&gt;READY&lt;/span&gt;
                &lt;/div&gt;
                &lt;ul className="mission-rail-list"&gt;
                    {MISSIONS.map((mission, index) => {
                        const isActive = mission.id === activeMission;
                        const isCompleted = index &lt; activeIndex;

                        return (
                            &lt;li
                                key={mission.id}
                                className={`mission-rail-item ${isActive ? "active" : ""} ${
                                    isCompleted ? "completed" : ""
                                }`}
                            &gt;
                                &lt;span className="rail-num"&gt;{mission.num}&lt;/span&gt;
                                &lt;span className="rail-label"&gt;{mission.label}&lt;/span&gt;
                                &lt;span className="rail-marker" /&gt;
                            &lt;/li&gt;
                        );
                    })}
                &lt;/ul&gt;
            &lt;/div&gt;
        &lt;/nav&gt;
    )}
    &lt;/&gt;
);

```

```

    }
    </ul>
    </div>
    </nav>
    </>
  })
  </>
};
}

```

File (22/30): components\PixelWorld.js

```

export function PixelWorld() {
  return (
    <div className="pixel-world" aria-hidden="true">
      <svg viewBox="0 0 960 420" role="presentation" shapeRendering="crispEdges">
        <rect className="pixel-sky" width="960" height="420" />
        <g className="pixel-stars">
          <rect x="64" y="38" width="8" height="8" />
          <rect x="156" y="78" width="6" height="6" />
          <rect x="742" y="42" width="8" height="8" />
          <rect x="852" y="92" width="6" height="6" />
        </g>
        <g className="pixel-clouds">
          <rect x="96" y="74" width="168" height="14" />
          <rect x="126" y="60" width="92" height="14" />
          <rect x="660" y="86" width="194" height="14" />
          <rect x="700" y="72" width="114" height="14" />
        </g>
        <g className="pixel-city-back">
          <rect x="0" y="180" width="110" height="200" />
          <rect x="126" y="222" width="112" height="158" />
          <rect x="254" y="154" width="144" height="226" />
          <rect x="414" y="206" width="118" height="174" />
          <rect x="548" y="136" width="164" height="244" />
          <rect x="728" y="196" width="104" height="184" />
          <rect x="848" y="164" width="112" height="216" />
        </g>
        <g className="pixel-windows">
          [
            [24, 214], [58, 214], [24, 248], [58, 282], [148, 248],
            [188, 248], [278, 188], [320, 188], [362, 222], [438, 238],
            [474, 238], [578, 172], [620, 172], [662, 206], [752, 228],
            [786, 262], [874, 198], [914, 232]
          ].map(([x, y]) => <rect key={` ${x} - ${y} `} x={x} y={y} width="16" height="10" />);
        </g>
        <g className="pixel-platforms">
          <rect x="0" y="350" width="244" height="18" />
          <rect x="98" y="330" width="188" height="12" />
          <rect x="336" y="310" width="246" height="20" />
          <rect x="382" y="288" width="154" height="12" />
          <rect x="670" y="338" width="290" height="20" />
          <rect x="730" y="316" width="170" height="12" />
        </g>
        <g className="pixel-circuit">
          <path d="M38 144H210V112H350V132H506" />
          <path d="M506 132H690V104H922" />
          <rect x="202" y="104" width="16" height="16" />
          <rect x="498" y="124" width="16" height="16" />
          <rect x="682" y="96" width="16" height="16" />
        </g>
        <g className="pixel-antenna">
          <rect x="810" y="154" width="12" height="184" />
          <rect x="786" y="150" width="60" height="12" />
          <rect x="796" y="122" width="40" height="28" />
          <rect x="804" y="110" width="24" height="12" />
        </g>
        <g className="pixel-bot">
          <rect x="146" y="288" width="52" height="42" />
          <rect x="154" y="276" width="36" height="12" />
          <rect x="138" y="298" width="8" height="20" />
          <rect x="198" y="298" width="8" height="20" />
          <rect x="154" y="330" width="10" height="18" />
          <rect x="180" y="330" width="10" height="18" />
          <rect className="pixel-bot-eye" x="158" y="296" width="8" height="8" />
        </g>
      </svg>
    </div>
  );
}

```

```

        <rect className="pixel-bot-eye" x="178" y="296" width="8" height="8" />
      </g>
    </svg>
    <span className="pixel-world-label">WORLD_01 / ENGINEERING DISTRICT</span>
  </div>
);
}

```

File (23/30): components\PostCard.js

```

import Link from "next/link";

function formatDate(value) {
  if (!value) return "";

  return new Intl.DateTimeFormat("en", {
    month: "short",
    day: "2-digit",
    year: "numeric",
    timeZone: "UTC"
  }).format(new Date(value));
}

export function PostCard({ post, index, featured = false }) {
  return (
    <article className={`post-card ${featured ? "featured" : ""}`>
      <div className="post-index mono">{String(index + 1).padStart(2, "0")}</div>
      <div className="post-main">
        <div className="post-meta">
          <span>{post.category}</span>
          <span>{formatDate(post.publishedAt)}</span>
          <span>{post.readingTime} min</span>
        </div>
        <h3>
          <Link href={`/writing/${post.slug}`}>{post.title}</Link>
        </h3>
        <p>{post.summary}</p>
        <div className="tag-row">
          {post.tags.slice(0, 4).map((tag) => (
            <span className="tag" key={tag}>
              {tag}
            </span>
          ))}
        </div>
      </div>
      <Link
        className="post-arrow"
        href={`/writing/${post.slug}`}
        aria-label={`Read ${post.title}`}
      >
        <span aria-hidden="true">▶</span>
      </Link>
    </article>
  );
}

```

File (24/30): components\ProjectCard.js

```

export function ProjectCard({ project, index, compact = false }) {
  return (
    <article
      className={`project-card ${compact ? "compact" : ""}`}
      id={project.slug}
    >
      <div className="project-card-top">
        <span className="mono muted">{String(index + 1).padStart(2, "0")}</span>
        <span className="mono muted">{project.year}</span>
      </div>

      {project.image ? (
        <div className="project-media">
          <img

```

```

        src={project.image}
        alt={`${project.title} hardware or interface`}
        loading="lazy"
      /&gt;
    &lt;/div&gt;
  ) : (
    &lt;div className="project-media project-placeholder" aria-hidden="true"&gt;
      &lt;span&gt;{project.tags[0]}&lt;/span&gt;
      &lt;span className="project-signal"&gt;■&lt;/span&gt;
    &lt;/div&gt;
  )}

{compact &amp;&amp; project.gallery?.length ? (
  &lt;div
    className="project-evidence-strip"
    aria-label={`${project.title} additional images`}
  &gt;
    {project.gallery.map((image) => (
      &lt;img src={image.src} alt={image.alt} loading="lazy" key={image.src} /&gt;
    ))}
  &lt;/div&gt;
) : null}
&lt;div className="project-copy"&gt;
  &lt;p className="project-category"&gt;{project.category}&lt;/p&gt;
  &lt;h3&gt;{project.title}&lt;/h3&gt;
  &lt;p&gt;{project.description}&lt;/p&gt;
&lt;/div&gt;

&lt;div className="project-meta"&gt;
  &lt;div className="tag-row"&gt;
    {project.tags.map((tag) => (
      &lt;span className="tag" key={tag}&gt;
        {tag}
      &lt;/span&gt;
    ))}
  &lt;/div&gt;
  &lt;span className="outcome"&gt;Result / {project.outcome}&lt;/span&gt;
&lt;/div&gt;
&lt;/article&gt;
);
}

```

File (25/30): components\SectionHeading.js

```

export function SectionHeading({ eyebrow, title, action }) {
  return (
    &lt;div className="section-heading"&gt;
      &lt;div&gt;
        {eyebrow ? &lt;p className="eyebrow"&gt;{eyebrow}&lt;/p&gt; : null}
        &lt;h2&gt;{title}&lt;/h2&gt;
      &lt;/div&gt;
      {action}
    &lt;/div&gt;
  );
}

```

File (26/30): components\SiteFooter.js

```

import Link from "next/link";
import { SocialIcon } from "@components/SocialIcon";
import { profile } from "@data/portfolio";

export function SiteFooter() {
  return (
    &lt;footer className="site-footer"&gt;
      &lt;div className="shell footer-grid"&gt;
        &lt;div&gt;
          &lt;p className="eyebrow"&gt;Embedded systems collaboration&lt;/p&gt;
          &lt;h2&gt;Need help with hardware, firmware, or a connected product?&lt;/h2&gt;
        &lt;/div&gt;
        &lt;div className="footer-contact"&gt;
          &lt;a className="text-link large" href={`mailto:${profile.email}`}&gt;

```

```

        Email Ahmed <span aria-hidden="true">&rarr;</span>
    </a>
    <div className="footer-links">
        {profile.socials.map((item) => (
            <a
                className="social-link"
                key={item.label}
                href={item.href}
                target="_blank"
                rel="noreferrer"
                aria-label={` ${item.label} profile (opens in a new tab)`}
            >
                <SocialIcon label={item.label} />
                <span>{item.label}</span>
            </a>
        ))}
    </div>
</div>
<div className="shell footer-bottom">
    <span>&copy; {new Date().getFullYear()} AHMED ASL</span>
    <span>EMBEDDED SYSTEMS & IOT & ROBOTICS</span>
    <Link href="/writing">READ ENGINEERING NOTES &rarr;</Link>
</div>
</footer>
);
}

```

File (27/30): components\SiteHeader.js

```

"use client";

import Link from "next/link";
import { usePathname } from "next/navigation";
import { useState } from "react";

const links = [
    { href: "/work", label: "01 / Work" },
    { href: "/about", label: "02 / About" },
    { href: "/writing", label: "03 / Writing" },
    { href: "/contact", label: "04 / Contact" }
];

export function SiteHeader() {
    const pathname = usePathname();
    const [open, setOpen] = useState(false);

    return (
        <header className="site-header">
            <div className="shell header-inner">
                <Link
                    className="brand"
                    href="/"
                    onClick={() => setOpen(false)}
                    aria-label="Ahmed Asl portfolio home"
                >
                    <span className="brand-mark" aria-hidden="true">A</span>
                    <span>AHMED ASL // EMBEDDED SYSTEMS</span>
                </Link>

                <button
                    className="menu-toggle"
                    type="button"
                    aria-expanded={open}
                    aria-controls="site-navigation"
                    onClick={() => setOpen((value) => !value)}
                >
                    <span>{open ? "Close menu" : "Open menu"}</span>
                    <span aria-hidden="true">{open ? "x" : "+"}</span>
                </button>

                <nav
                    id="site-navigation"
                    className={`site-nav ${open ? "is-open" : ""}`}
                    aria-label="Primary navigation"
                >

```

```

    &gt;
    {links.map((link) => (
      <Link
        key={link.href}
        href={link.href}
        className={pathname.startsWith(link.href) ? "active" : ""}
        aria-current={pathname.startsWith(link.href) ? "page" : undefined}
        onClick={() => setOpen(false)}
      >
        {link.label}
      </Link>
    ))}
    </nav>
  </div>
</header>
);
}

```

File (28/30): components\SocialIcon.js

```

import { faLinkedin } from "@fortawesome/free-brands-svg-icons";
import {
  siBehance,
  siGithub,
  siGooglescholar,
  siTryhackme,
  siYoutube
} from "simple-icons";

const icons = {
  GitHub: { path: siGithub.path, viewBox: "0 0 24 24" },
  LinkedIn: {
    path: faLinkedin.icon[4],
    viewBox: `0 0 ${faLinkedin.icon[0]} ${faLinkedin.icon[1]}`
  },
  "Google Scholar": { path: siGooglescholar.path, viewBox: "0 0 24 24" },
  TryHackMe: { path: siTryhackme.path, viewBox: "0 0 24 24" },
  Behance: { path: siBehance.path, viewBox: "0 0 24 24" },
  YouTube: { path: siYoutube.path, viewBox: "0 0 24 24" }
};

export function SocialIcon({ label }) {
  const icon = icons[label];
  if (!icon) return null;

  return (
    <svg
      className="social-icon"
      viewBox={icon.viewBox}
      aria-hidden="true"
      focusable="false"
    >
      <path d={icon.path} />
    </svg>
  );
}

```

File (29/30): components\WorldGallery.js

```

"use client";

import Link from "next/link";
import { useEffect, useState } from "react";
import { ambientWorlds } from "@/data/ambient";

export function WorldGallery() {
  const [animationsEnabled, setAnimationsEnabled] = useState(false);

  useEffect(() => {
    const mediaQuery = window.matchMedia("(prefers-reduced-motion: reduce)");
    const syncPreference = () => setAnimationsEnabled(!mediaQuery.matches);
  });
}

```

```

syncPreference();
mediaQuery.addEventListener?.("change", syncPreference);
return () => mediaQuery.removeEventListener?.("change", syncPreference);
}, []);

return (
  <section className="section world-select-section" aria-labelledby="world-select-title">
    <div className="shell">
      <div className="world-select-heading">
        <div>
          <p className="eyebrow">Visual index / Topics</p>
          <h2 id="world-select-title">Choose a topic</h2>
          <p className="world-select-intro">
            Each scene marks a subject area. Select one to browse those field notes.
          </p>
        </div>
      </div>

      <div className="world-grid">
        {ambientWorlds.map((world, index) => (
          <Link
            className={`world-card${world.featured ? " featured" : ""}`}
            key={world.id}
            href={`/writing?topic=${world.filterTerm}#published-field-logs`}
            aria-label={`Browse ${world.label} articles`}
          >
            <div className="world-card-top mono">
              <span>WORLD_{String(index + 1).padStart(2, "0")}</span>
              <span>{world.label}</span>
            </div>
            <div className="world-media">
              {animationsEnabled ? (
                <img
                  src={world.image}
                  alt={`Pixel-art scene representing ${world.label}`}
                  loading={world.featured ? "eager" : "lazy"}
                  referrerPolicy="no-referrer"
                />
              ) : (
                <div
                  className="world-paused"
                  role="img"
                  aria-label={` ${world.label} topic`}
                >
                  <span className="world-paused-icon" aria-hidden="true">II</span>
                  <span className="mono">REDUCED MOTION</span>
                </div>
              )}
            </div>
            <div className="world-card-copy">
              <div>
                <h3>{world.title}</h3>
                <p>{world.description}</p>
              </div>
              <span className="text-link" aria-hidden="true">
                Browse notes <span aria-hidden="true">■</span>
              </span>
            </div>
          </Link>
        ))}
      </div>
    </div>
  </section>
);
}

```

File (30/30): components\WritingIndex.js

```

"use client";

import { useMemo, useState, useEffect } from "react";
import { PostCard } from "../PostCard";

export function WritingIndex({ posts }) {
  const [query, setQuery] = useState("");

```

```

const [category, setCategory] = useState("All");
const categories = ["All", ...new Set(posts.map((post) => post.category))];

// Initialize query from URL search params (e.g. ?topic=linux)
useEffect(() => {
  if (typeof window !== "undefined") {
    const params = new URLSearchParams(window.location.search);
    const topic = params.get("topic");
    if (topic) {
      setQuery(topic);
      // Scroll to the index section if a topic was selected
      const element = document.getElementById("published-field-logs");
      if (element) {
        setTimeout(() => element.scrollIntoView({ behavior: "smooth" }), 100);
      }
    }
  }
}, []);

const filtered = useMemo(() => {
  const needle = query.trim().toLowerCase();
  return posts.filter((post) => {
    const inCategory = category === "All" || post.category === category;
    const haystack = [
      post.title,
      post.summary,
      post.category,
      ...post.tags
    ]
      .join(" ")
      .toLowerCase();
    return inCategory &amp; (!needle || haystack.includes(needle));
  });
}, [posts, query, category]);

return (
  <>
    <div className="writing-tools">
      <label className="search-field">
        <span className="sr-only">Search published engineering articles</span>
        <span className="mono" aria-hidden="true">SEARCH</span>
        <input
          value={query}
          onChange={(event) => setQuery(event.target.value)}
          placeholder="Search by topic, tool, or command"
        />
      </label>
      <div className="filter-row" aria-label="Filter articles by category">
        {categories.map((item) => (
          <button
            className={category === item ? "active" : ""}
            key={item}
            type="button"
            aria-pressed={category === item}
            onClick={() => setCategory(item)}
          />
          {item}
        </button>
        ))}
      </div>
    </div>

    <div className="post-list">
      {filtered.map((post, index) => (
        <PostCard key={post.slug} post={post} index={index} />
      ))}
    </div>

    {!filtered.length ? (
      <div className="empty-state">
        <p className="eyebrow">No matching articles</p>
        <h2>Change the search term or choose another category.</h2>
      </div>
    ) : null}
  </>
);

```

